

Upstream, downstream, and in between: Navigating the GPAI value chain under EU Law

By Martin Ebers* and Emmanuel Vargas Penagos*

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* President of the Robotics & AI Law Society (RAILS), WASP-HS Guestprofessor at Örebro University, Sweden.

* PhD student, School of Behavioural, Social and Legal Sciences, Örebro University, Sweden.

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Abstract

This article examines how responsibility and liability are allocated across the general-purpose AI (GPAI) value chain under EU law. Recognising the fluid and dynamic nature of this value chain, we analyse the obligations imposed on upstream and downstream providers in the AI Act, as well as in complementary instruments such as the GPAI Code of Practice and the European Commission's Guidelines on the scope of the obligations for GPAI models. Our paper identifies gaps in these instruments and explores how they intersect with obligations under the General Product Safety Regulation (GPSR) and the new Product Liability Directive (PLD). The article also considers key aspects of model deployment and distribution under the Digital Services Act (DSA), particularly in relation to GPAI marketplaces and very large online platforms (VLOPs). It highlights the central role of cooperation duties in effective risk mitigation, while emphasizing the asymmetries of power and information that may hinder such cooperation. Against this background, the article assesses where responsibility is most appropriately allocated along the GPAI value chain.

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1. Introduction

1.1. Risks and Opportunities of General Purpose AI

In early 2022, a group of researchers who had developed MegaSyn, an AI model for drug discovery, found that, with relatively simple modifications, the system could generate 40,000 potentially lethal molecules, some of which resembled VX, one of the most potent chemical warfare agents in history.¹ Months later, following the proliferation of ChatGPT among the general public, researchers raised concerns about the potential of Large Language Models (LLM) to assist in the ideation, planning and execution of the design of biological weapons.² These risks, alongside other risks emerging from misuse and malfunction, as well as systemic risks that can originate from the widespread deployment of general-purpose AI (GPAI), were highlighted in the International Scientific Report on the Safety of Advanced AI of January 2025 as issues that need to be addressed to enable society to enjoy the full benefits of AI. As the report noted, GPAI's capacity to perform a broad range of indeterminate activities can have a wide range of dual-use applications that can benefit or harm of people. For example, it can be used to write computer code and to accelerate research in several scientific fields like medicine and pharmaceuticals.³ However, it can also be used to enhance the capacities of malicious actors to conduct cyber-attacks or create non-consensual 'deepfake' pornography.⁴

GPAI models, also known as foundation models in technical research, can be considered to be 'the prior or the foundation of one or several other models' by fine-tuning⁵ or adapting them to specific tasks.⁶ Furthermore, given their general nature and the implementation of user-friendly interfaces, several of these models can be used for various activities without any further development and without a need for the users to be computer scientists or engineers. Instead, users can apply 'prompt engineering' strategies.⁷ Research in this field has focused on demonstrating how generative AI can be introduced into various types of services, such as business analytics for predicting customer behaviour, identifying fraud and providing user recommendations, by 'crafting specific inputs to guide the AI models towards generating relevant and accurate outputs'.⁸ Healthcare is another sector developing prompt engineering

¹ Fabio Urbina and others, 'Dual Use of Artificial-Intelligence-Powered Drug Discovery' (2022) 4 Nature Machine Intelligence 189.

² Jonas B Sandbrink, 'Artificial Intelligence and Biological Misuse: Differentiating Risks of Language Models and Biological Design Tools' (arXiv, 2023) <<https://arxiv.org/abs/2306.13952>> accessed 30 April 2025.

³ 'Science in the Age of AI: How Artificial Intelligence Is Changing the Nature and Method of Scientific Research' (The Royal Society) <<https://royalsociety.org/-/media/policy/projects/science-in-the-age-of-ai/science-in-the-age-of-ai-report.pdf#page=6.26>> accessed 25 August 2025.

⁴ AI Action Summit, 'International AI Safety Report' <https://assets.publishing.service.gov.uk/media/679a0c48a77d250007d313ee/International_AI_Safety_Report_2025_accessible_f.pdf>.

⁵ D Diffan Nur Cahyo, Andi Sunyoto and Dhani Ariatmanto, 'Transfer Learning and Fine-Tuning Effect Analysis on Classification of Cat Breeds Using a Convolutional Neural Network', *2023 6th International Conference on Information and Communications Technology (ICOIACT)* (IEEE 2023) <<https://ieeexplore.ieee.org/document/10455771/>> accessed 7 April 2025.

⁶ Fine-tuning can be understood as 'the process of continuing training or adjusting existing models with smaller specific datasets or different domains.' Ali Hussain and others, 'Foundation Models: From Current Developments, Challenges, and Risks to Future Opportunities', *2025 27th International Conference on Advanced Communications Technology (ICACT)* (2025) <<https://ieeexplore.ieee.org/document/10936649>> accessed 22 April 2025.

⁷ Rushali Deshmukh and others, 'Optimizing Human-AI Interaction: Innovations in Prompt Engineering', *2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)* (IEEE 2025) <<https://ieeexplore.ieee.org/document/10914815/>> accessed 23 April 2025.

⁸ Fatima Qiyam, Yousef Jaradat and Shadi AlZu'Bi, 'Prompt Engineering in Business Analytics for Next-Gen Digital Banking Services', *2024 2nd International Conference on Foundation and Large Language Models (FLLM)* (2024) <<https://ieeexplore.ieee.org/document/10852452>> accessed 9 May 2025.

strategies, with relevant advances in its use for diagnosis and decision-making systems.⁹ These models have the capacity to be deployed across a wide range of industries, activities, and markets.

1.2. The GPAI value chain

Beyond the opportunities to expand AI's potential uses for the benefit of users, the variety of possible downstream applications also creates regulatory gaps, overlaps and uncertainties. These possibilities also give rise to risks and compliance challenges. For instance, when a model is adapted for the use in a specific sector, it may become subject to sector-specific legal requirements – such as health-sector legislation applicable to an LLM fine-tuned for medical services.¹⁰ Likewise, technical research has shown that the flexibility of AI-human interaction through prompts introduces additional safety and compliance risks. Different prompting techniques, broadly referred to as 'prompt hacking', can be exploited to induce outputs that developers initially sought to prevent, sometimes with harmful consequences¹¹ and often undermining the efforts of providers to ensure legal compliance.¹²

Furthermore, most generative GPAI-based systems rely on user inputs or prompts to generate responses. This carries the risk that the generated content could be harmful or illegal as a result of the prompt applied by the user. To address this issue, developers implement 'prompt moderation' strategies. Some of these strategies involve analysing and filtering prompts before the AI generates any content,¹³ while others involve interfering with the output, for example by blurring or pixelating it.¹⁴ These strategies can be implemented at different stages of development, including initial model creation, modification processes, and integration into a system.

Approaching these challenges from a regulatory perspective is complex, in part due to the diverse range of stakeholders involved in the development and use of GPAI models. The GPAI value chain comprises several tiers of stakeholders who have varying degrees of influence over the capabilities of this technology for better or worse. This also means that the way in which the responsibility and potential liability for misuse or malfunction can easily shift from one element of the value chain to another.

1.3. Regulatory responses in the EU

These circumstances have attracted increasing attention from legislators, policymakers, human rights bodies, civil society and academia worldwide. Against this backdrop, the EU

⁹ Tharindu Weerathunge, Shantha Jayalal and Keerthi Wijayasiriwardhane, 'Optimizing Response Consistency of Large Language Models in Medical Education through Prompt Engineering', *2025 5th International Conference on Advanced Research in Computing (ICARC)* (IEEE 2025) <<https://ieeexplore.ieee.org/document/10963313/>> accessed 9 May 2025.

¹⁰ Hussain and others (n 6).

¹¹ Baha Rababah and others, 'SoK: Prompt Hacking of Large Language Models', *2024 IEEE International Conference on Big Data (BigData)* (IEEE 2024) <<https://ieeexplore.ieee.org/document/10825103/>> accessed 23 April 2025.

¹² Oskar J Gstrein, Noman Haleem and Andrej Zwitter, 'General-Purpose AI Regulation and the European Union AI Act' (2024)

¹³ Internet Policy Review <<https://policyreview.info/articles/analysis/general-purpose-ai-regulation-and-ai-act>> accessed 23 April 2025.

¹⁴ Praful Pardhi, 'Content Moderation of Generative AI Prompts' (2025) 6 SN Computer Science 329.

¹⁴ Piera Riccio, Georgina Curto and Nuria Oliver, 'Exploring the Boundaries of Content Moderation in Text-to-Image Generation' (arXiv, 2024) <<https://arxiv.org/abs/2409.17155>> accessed 20 May 2025.

legislature has introduced the Artificial Intelligence Act (AI Act)¹⁵ establishing a framework of obligations for stakeholders across the AI value chain, including specific requirements for those engaged in developing and using GPAI technology.

For methodological clarity, it is important to note that the AI Act places the most stringent regulatory requirements on high-risk AI systems, rather than on the models themselves. Although the AI Act does not explicitly define the term ‘AI model’,¹⁶ it consistently distinguishes between models and systems. Recital 97 clarifies that AI models—such as GPAI models—may constitute essential components of AI systems, but do not, on their own, amount to systems. Incorporating a model into a system typically requires additional elements, such as a user interface or application layer, to enable interaction with the model's outputs. The European Commission's Guidelines on the scope of the obligations for general-purpose AI models adopt this approach. In accordance with Recital 101 of the AI Act, they note that GPAI model providers have a particular role and responsibility along the value chain to provide information that enables a ‘good understanding’ of the model and its capabilities, which is necessary for both integrating models within systems and for complying with the AI Act requirements.¹⁷

Against this background, Table 1 provides a non-exhaustive overview of the economic operators specifically involved in the GPAI-value chain, situating them within the AI Act, the General Product Safety Regulation (GPSR),¹⁸ the new Product Liability Directive (PLD),¹⁹ and the Digital Services Act (DSA),²⁰ where applicable. However, it should be noted that not all operators in the value chain (e.g. importers) are covered.

Table 1: GPAI-value chain and the place of its economic operators in the relevant EU legal framework²¹					
Stakeholder	Role	AI Act	GPSR	PLD	DSA
Compute and cloud providers	Providers of computing infrastructure	Not regulated	Not regulated	‘Related service’ (Art 4(3))	‘Hosting’ (Art 3(g)(iii))

¹⁵ Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 [2024] OJ L, 2024/1689, 12.7.2024 (AI Act).

¹⁶ ISO/IEC 22989 standard defines ‘model’ as a ‘physical, mathematical or otherwise logical representation of a system, entity, phenomenon, process or data’.

¹⁷ Commission, ‘ANNEX to the Communication to the Commission Approval of the content of the draft Communication from the Commission – Guidelines on the scope of the obligations for general-purpose AI models established by Regulation (EU) 2024/1689 (AI Act) 2025’ C(2025) 5045 final, para 2.

¹⁸ Regulation (EU) 2023/988 of the European Parliament and of the Council of 10 May 2023 on general product safety, amending Regulation (EU) No 1025/2012 of the European Parliament and of the Council and Directive (EU) 2020/1828 of the European Parliament and the Council, and repealing Directive 2001/95/EC of the European Parliament and of the Council and Council Directive 87/357/EEC [2023] OJ L 135, 23.5.2023, pp. 1–51 (GPSR).

¹⁹ Directive (EU) 2024/2853 of the European Parliament and of the Council of 23 October 2024 on liability for defective products and repealing Council Directive 85/374/EEC [2024] OJ L, 2024/2853, 18.11.2024 (PLD).

²⁰ Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC [2022] OJ L 277, 27.10.2022, pp. 1–102 (DSA).

²¹ Authors’ own work based on the table created by Madhulika Srikumar, Jiyou Chang and Kasia Chmielinski, ‘Risk Mitigation Strategies for the Open Foundation Model Value Chain’ (*Partnership on AI*, 11 July 2024) <<https://partnershiponai.org/resource/risk-mitigation-strategies-for-the-open-foundation-model-value-chain/>> accessed 16 June 2025.

	for training and running models.				
Data providers	Providers of datasets necessary for the training or fine-tuning of models.	Not regulated	Unclear (potentially falling within 'Manufacturer' (Art 3(8)) of a 'component')	Potentially 'Manufacturer' (Art 4(10)) of a 'related service', which in turn is also considered as a 'component' (Art 4(4)), (Art 4(15))	Not regulated
GPAI Model providers	Initial model developers	'Provider' (Art 3(3))	'Manufacturer' (Art 3(8))	'Manufacturer' (Art 4(10))	Not regulated
GPAI downstream modifiers	Developers involved in fine-tuning or adapting models for specific uses	Unclear (potentially falling within 'Provider' (Art 3(3))	'Manufacturer' (Art 3(8))	'Manufacturer' (Art 4(10)) in case of a 'substantial modification' (Art 4(18))	Not regulated
Model marketplaces	Platforms making models, adapted models, and other components available to other developers	Not regulated	'Distributor' (Art 4(11))	'Distributor' (Art 4(14))	'Hosting' (Art 3(g)(iii)), 'Online platform' (Art 3(i))
App developers, service developers, model integrators	Developers integrating models in their AI systems (e.g. chatbots, models available via API, digital assistants)	'Downstream provider' (Art 3(68))	'Manufacturer' (Art 3(8))	'Manufacturer' (Art 4(10))	In some cases, 'Online platform' (Art 3(i), 'Online search engine' (Art 3(j))
Distribution platforms	Platforms in which applications and content are distributed	'Provider' (Art 3(3)), 'Distributor' (Art 3(7)),	'Distributor' (Art 3(11)).	'Distributor' (Art 4(14)).	In some cases, 'Online platform' (Art 3(i), 'Online

	(e.g. App Stores)				search engine' (Art 3(j))
	Platforms in which applications and content are used (e.g. social media platforms)	'Deployer' (Art 3(4))	Not addressed in the GPSR ²²	Not addressed in the PLD. ²³ However, if the platforms incorporate distinguishing features on the service that makes them seem like a manufacturer, they can turn into an apparent manufacturer (Art 4(10)(b))	In some cases, 'Online platform' (Art 3(i), 'Online search engine' (Art 3(j))
Users	Users at a professional capacity	'Deployer' (Art 3(4)) The AI Act refers to use under their authority. Deployers will typically deploy 'staff' (undefined by the AI Act) to operate the AI system on their behalf	Not addressed in the GPSR due to the Regulation's scope on 'consumer' products ²⁴	Not addressed in the PLD as it only covers damage emerging from non-professional activities and to damage to data not used in commercial activities ²⁵	'Recipient of the service' (Art 3(b)) or 'Trader' (Art 3(f))
	Users at a non-professional capacity	Excluded from the AI Act (Art 3(4))	'Consumer' (Art 3(17)) in the case of natural persons	'Consumer' - undefined but set in the Directive's scope (Art 1)	'Recipient of the service' (Art 3(b)) or 'Consumer' (Art 3(c))

²² In this case, platforms embedding the model within their services and for the use of their users would be using it as a service, which are excluded from the GPSR. See GPSR, Recital 17.

²³ In this case, platforms embedding the model within their services and for the use of their users would be using it as a service, which are excluded from the PLD. See PLD, Recital 17.

²⁴ The GPSR is only applicable to 'consumer products' and 'consumer' is a natural person acting in non-professional settings. See GPSR, arts 1(2) and 3(17).

²⁵ The liability regime under the PLD is only triggered under non-professional activities. See PLD, art 6(1)(b)(iii) and 6(1)(c).

1.4. The dynamic nature of GPAI-value chains

The GPAI value chain is not necessarily linear. Activities may occur in parallel. They may recur at different stages. Or they may unfold in varying sequences. Moreover, several actors may participate in more than one layer of the value chain, or even all of them.²⁶ Rather than resembling a traditional industrial supply chain, the GPAI value chain is better understood as a network of ‘cocreation relations’ through which resources flow downstream, upstream, and horizontally.²⁷

A useful metaphor is that of a river. While it may appear to consist of a single, unified stream, a river actually contains multiple currents moving at different speeds and in different directions. While the main current flows downstream, smaller currents may flow upstream, and eddies or whirlpools emerge where these opposing streams interact.²⁸ Figure 1 below illustrates how such dynamics can manifest in practice.

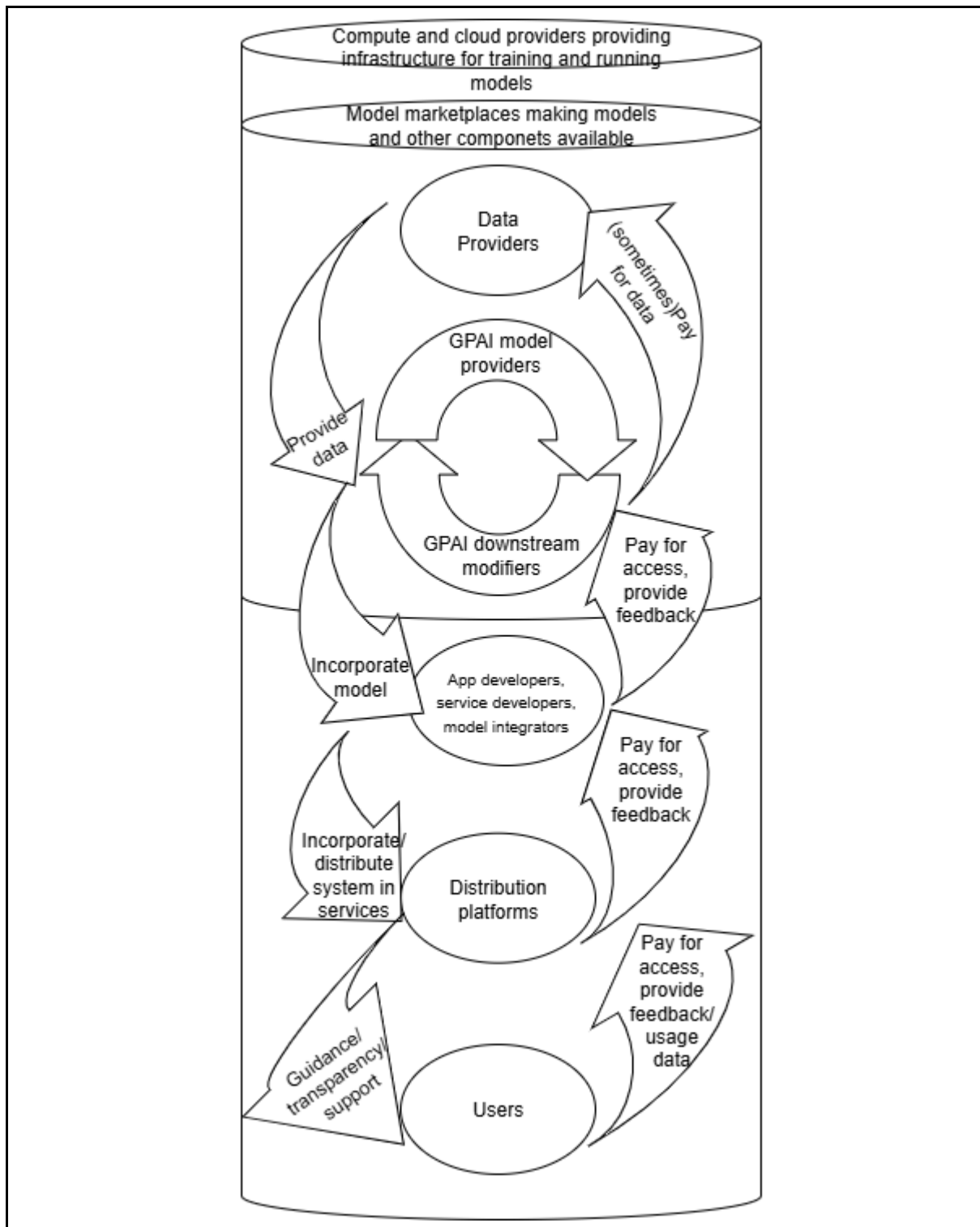
Figure 1: Illustration of the GPAI value chain dynamics²⁹

²⁶ Madhulika Srikumar, Jiyoo Chang and Kasia Chmielinski, ‘Risk Mitigation Strategies for the Open Foundation Model Value Chain’ (*Partnership on AI*, 11 July 2024).

²⁷ Blair Attard-Frost and David Gray Widder, ‘The Ethics of AI Value Chains’ (2025) 12 *Big Data & Society* <<https://journals.sagepub.com/doi/10.1177/20539517251340603>> accessed 2 July 2025.

²⁸ Ellie Friedmann, ‘What Are the Different Types of River Currents?’ (*Whitewater Guidebook*, 22 May 2020) <<https://www.whitewaterguidebook.com/what-are-the-different-types-of-river-currents/>> accessed 1 July 2025.

²⁹ Authors’ own work based on the table created by Srikumar, Chang and Chmielinski (n 26) and the figure created by Attard-Frost and Widder (n 27).



The way in which this flow is configured in practice may vary depending on several factors, including whether a stakeholder is involved in more than one level of the value chain. As Engler and Renda point out, there are various ways in which the value chain can be structured at a commercial level, with stakeholders likely to engage in business models involving different layers of the value chain. The authors argue that the AI value chain is an organisational process through which AI systems are developed and deployed, involving several stages. They further argue that these stages create break points at which different actors can

contribute in different ways, which makes the value chain dynamic and varied across systems and actors. This is essential for allocating regulatory responsibilities.³⁰

1.5. Scope and structure of this article

Building on this background, this article analyses how the various economic operators involved in the GPAI value chain—from the initial development stage onwards—are covered by existing EU secondary legislation on responsibility and liability. The analysis begins with the AI Act and its obligations for the development stages of the GPAI value chain. First, it considers the role of initial model developers or *upstream providers* (2.). Next, it turns to *downstream providers* and *modifiers* (3.). Finally, it addresses the cooperation duties between upstream and downstream providers (4.).

The discussion then shifts to the regulatory requirements arising from product safety and liability legislation (5.). Thereafter, the paper examines the applicable regime for GPAI model distribution platforms, also known as ‘model marketplaces’ (6.) before turning platforms classified as Very Large Online Platforms (VLOPs) under the DSA that integrate GPAI technologies into their services (7.). The analysis is followed by an evaluative section, which considers where responsibility is most appropriately allocated within the GPAI value chain (8.). Finally, the article concludes by drawing together the findings and outlining their broader implications for the regulation of GPAI (9.).

Outside of the scope are unresolved issues such as the specific measures that providers should adopt to mitigate risks,³¹ how chatbots should communicate with users,³² and under what circumstances liability may ultimately rest with the end user. Moreover, this article does not address issues related to enforcement and market surveillance. Rules established in the Digital Markets Act³³ and competition law in general, which may be relevant in cases in which stakeholders in the value chain are in a dominant position, are also outside of the scope of this article.

³⁰ Alex Engler and Andrea Renda, ‘Reconciling the AI Value Chain with the EU’s Artificial Intelligence Act’ (CEPS).

³¹ An example of this can be seen in the discussions emerging from ChatGPT’s ‘sycophancy’ in responding to user queries. See: ‘Sycophancy in GPT-4o: What Happened and What We’re Doing about It’ <<https://openai.com/index/sycophancy-in-gpt-4o/>> accessed 25 August 2025.

³² Several instances about this issue have emerged in the last year. See for instance: ‘Meta’s AI Rules Have Let Bots Hold “Sensual” Chats with Children’ *Reuters* (14 August 2025) <<https://www.reuters.com/investigates/special-report/meta-ai-chatbot-guidelines/>> accessed 25 August 2025.

³³ Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector and amending Directives (EU) 2019/1937 and (EU) 2020/1828 [2022] OJ L 265, 12.10.2022, pp. 1–66.

2. Obligations of GPAI model providers under the AI Act

2.1. Overview

During the legislative process leading to the adoption of the AI Act, the co-legislators acknowledged the ability of AI systems—particularly tools like ChatGPT which are based on large language models (LLMs)—to perform a broad range of downstream tasks. In response, they introduced general-purpose AI (GPAI) as a distinct regulatory category within the Act.³⁴ As a result, Chapter V of the AI Act establishes a set of basic obligations for providers of GPAI models,³⁵ alongside stricter requirements for GPAI models deemed to present “systemic risk.”³⁶ A model falls into this latter category when it exceeds the defined technical threshold of 10^{25} floating-point operations used in training.³⁷ This category covers several popular LLMs that are widely available, such as OpenAI’s GPT-4, GPT-4o and GPT-4.5, Google’s Gemini Ultra, Xai’s Grok 3 and Grok 4, Meta’s Llama 3.1-405B, Mistral Large 2, and Claude 3 Plus.³⁸

As will be discussed below, the co-legislators opted to assign the majority of the safety-related obligations to providers of models with systemic risk.

The first instrument stemming from the AI Act to concretise companies’ obligations is the Code of Practice on general-purpose AI models. Codes of Practice are ‘encouraged’ under Article 56 of the AI Act and enable signatories to demonstrate to the Commission that they are fulfilling their duties under Articles 53 and 55. In August 2025, the Commission and the AI Board approved the final version of the GPAI code of practice, following a multistakeholder process that lasted over nine months. Comprising three chapters dedicated to transparency, copyright, safety and security, it is complemented by non-binding guidelines issued by the Commission.³⁹ While companies may, in principle, design alternative mechanisms to prove compliance, this would be far more complex, costly, and legally uncertain. The Code is therefore not merely a non-binding guideline but a tool with both practical and legal implications. It is also significant for downstream providers because it shapes what they can reasonably expect from GPAI models, influences the negotiation of contracts with GPAI providers, and specifies the information and documentation obligations owed to downstream providers under Article 53(1)(b) of the AI Act.

2.2. Obligations of providers of GPAI models

Article 53 of the AI Act sets out four transparency obligations for GPAI model providers. They must prepare technical documentation for supervisory authorities in accordance with the requirements detailed in Annex XI (point (a)); provide the necessary information to enable

³⁴ David Fernández-Llorca and others, ‘An Interdisciplinary Account of the Terminological Choices by EU Policymakers Ahead of the Final Agreement on the AI Act: AI System, General Purpose AI System, Foundation Model, and Generative AI’ [2024] Artificial Intelligence and Law <<https://link.springer.com/10.1007/s10506-024-09412-y>> accessed 4 February 2025.

³⁵ AI Act, art 53.

³⁶ *ibid*, art 55.

³⁷ *ibid*, art 55(2).

³⁸ ‘Data on AI Models’ (*Epoch AI*, 19 June 2024) <<https://epoch.ai/data/ai-models>> accessed 1 September 2025.

³⁹ ‘The General-Purpose AI Code of Practice | Shaping Europe’s Digital Future’ <<https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>> accessed 5 August 2025.

downstream compliance when integrating a GPAI model into their systems (point (b)); implement a copyright policy (point (c)); and publish a ‘sufficiently detailed summary about the content used for training’ the model, using a template issued by the Commission in July 2025 (point (d)).⁴⁰ However, all of these obligations, except for the copyright policy, are not applicable to providers of free and open source models.

The transparency chapter of the Code of Practice introduces a Model Documentation form for GPAI providers. The form captures the information required under Article 53(1)(a) and (b) to respond to requests from the AI Office and national authorities. It also allows downstream providers to integrate the models into their systems.⁴¹ In addition to the general information to identify the model, the form requires details on the model’s properties, methods of distribution and licenses, technical requirements for its use, training process, and the data used for its training, testing and validation. It also requires information on the computational resources and energy consumption during the training and inference phases.

In addition to the above, the transparency chapter is complemented by a template prepared by the AI Office to help GPAI providers to comply with their obligation under Article 53(1)(d) to draft a summary of the content used to train their model. The explanatory note to this template, issued in July 2025, makes direct reference to Recital 107 of the AI Act, which states that the information to be included in this document should be ‘generally comprehensive in its scope rather than technically detailed, to enable parties with legitimate interests, including copyright holders, to exercise and enforce their rights under Union law’. Based on this rationale, the template requires the provider and the model to be identified, and details to be provided on the modalities, size, type, language, and collection period of the training data. Providers must also disclose the sources of the data and describe the measures taken to respect copyright reservations under the text and data mining exception to copyright,⁴² as well as the measures taken to avoid or remove illegal content from the training data.

The copyright chapter of the Code of Practice focuses on the obligation of GPAI providers to create and implement a copyright policy. The measures set out in this chapter can be summarised as follows: first, to ensure that their web-crawler data mining does not infringe copyright by scraping or compiling data; and second, to mitigate copyright infringement by downstream AI systems by adopting technical measures to prevent the generation of outputs that reproduce copyright-protected content in an infringing manner, and to prohibit infringing uses in their acceptable use policy or similar documents.⁴³

⁴⁰ Commission ‘ANNEX to the Communication to the Commission Approval of the content of the draft Communication from the Commission – Explanatory Notice and Template for the Public Summary of Training Content for general-purpose AI models required by Article 53 (1)(d) of Regulation (EU) 2024/1689 (AI Act)’ C(2025) 5235 final.

⁴¹ Nuria Oliver (Working Group 1 Co-Chair) and Rishi Bommasani (Working Group 1 Vice-Chair), ‘Code of Practice for General-Purpose AI Models - Transparency Chapter’ (2025) <<https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>>.

⁴² Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market and amending Directives 96/9/EC and 2001/29/EC [2019] OJ L 130, 17.5.2019, pp. 92–125, art 4(3).

⁴³ Alexander Peukert (Working Group 1 Co-Chair) and Céline Castets-Renard (Working Group 1 Vice-Chair), ‘Code of Practice for General-Purpose AI Models - Copyright Chapter’ (2025) <<https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>>.

2.3. Obligations of providers of GPAI Models with systemic risks

Stricter obligations apply to GPAI models with systemic risks. According to Article 55 of the AI Act, providers of such models must, in addition to the obligations in Article 53: i) perform model evaluations with standardised state of the art protocols to identify and mitigate systemic risks; ii) assess and mitigate systemic risks at EU level, 'including their sources' emerging from the deployment, the placing on the market and the use of their models; iii) track, document and report to the AI Office and national authorities on 'serious incidents'⁴⁴ and possible corrective measures to address them; and iv) ensure cybersecurity protection for the model and its physical infrastructure.

The Safety and Security chapter of the Code of Practice⁴⁵ establishes a set of principles and commitments designed to guide providers of GPAI models with systemic risks in complying with the obligations set out in Article 55(1) of the AI Act. To this end, signatories must continuously assess and mitigate systemic risks, taking measures 'along the entire model lifecycle' in cooperation with all relevant stakeholders in the AI value chain.⁴⁶ This materialises in the form of a series of measures, including the creation and regular updating of a safety framework incorporating risk assessment and mitigation processes, the identification and modelling of systemic risks, and the analysis of their probability and severity, as well as post-market monitoring. Providers must establish criteria for risk acceptance and adopt mitigation and cybersecurity measures. They must also assign clear responsibilities and allocate resources within their organisations. They must also create processes and measures for documenting and reporting serious incidents to national authorities and the AI Office, as well as publishing summarised versions of their safety frameworks and reports.

Appendix 1 of this Chapter provides a clearer definition of the types of systemic risk covered, referring directly to Article 3(65), which lists risks to public health, safety, public security, fundamental rights, and society at large. It highlights, for example, risks to rights such as freedom of expression and information, non-discrimination, privacy, and data protection, as well as risks to democratic processes. It also identifies harmful content, including illegal, violent, hateful, radicalising, or false material, such as child sexual abuse material and non-consensual intimate images. To facilitate identification, the Appendix outlines criteria relating to the nature and sources of risks, such as: capabilities (e.g. manipulation, persuasion, deception, or multimodal processing), propensities (e.g. misalignment with human intent or values, harmful deployment of capabilities, or hallucinations), and affordances (e.g. model autonomy, susceptibility to adversarial removal of guardrails, interaction with other systems, or inappropriate uses). On this basis, the Appendix designates as 'specified systemic risks' those that may enable chemical, biological, radiological, or nuclear incidents; lead to loss of human control over directing, modifying, or shutting down a model; enable large-scale or sophisticated cyber-attacks; or facilitate harmful manipulation of human behaviour.

⁴⁴ AI Act, art 3(49) defines 'serious incident' as incidents or malfunctioning that have a direct or indirect result of an individual's death or serious harm, a serious and irreversible disruption for managing or operating critical infrastructure, the infringement of EU law obligations on the protection of fundamental rights, or serious harm to property or the environment. See also 'AI Act: Commission Issues Draft Guidance and Reporting Template on Serious AI Incidents, and Seeks Stakeholders' Feedback | Shaping Europe's Digital Future' <<https://digital-strategy.ec.europa.eu/en/consultations/ai-act-commission-issues-draft-guidance-and-reporting-template-serious-ai-incidents-and-seeks>> accessed 17 October 2025.

⁴⁵ Matthias Samwald (Working Group 2 Co-Chair) and others, 'Code of Practice for General-Purpose AI Models - Safety and Security Chapter' (2025) <<https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>>.

⁴⁶ *ibid.*, Principle a.

Commitment 5 on safety mitigations further specifies the measures that providers should consider. These include filtering and cleaning training data, monitoring and filtering inputs and outputs, modifying model behaviour, restricting API access to vetted users, and offering tools that enable downstream mitigation. However, the Code does not clearly explain how these techniques should be implemented in practice, nor does it clarify whether they can provide adequate protection against risks.

The interim conclusion is that systemic model providers face a wide range of responsibilities, but effective assessment and mitigation depend heavily on self-assessment and the discretion of providers. The Code leaves unresolved questions on how cooperation duties and safety requirements should be operationalised and only applies mitigation measures to a limited number of models that qualify as systemic risk. This leaves gaps with respect to GPAI systems more broadly. Similarly, as will be explained in the following section, GPAI system providers have no specific obligations under the AI Act.

3. Obligations of downstream providers under the AI Act

According to the AI Act ‘downstream providers’ are those who provide an AI system of a specific or general purpose which integrates an AI model, regardless of whether it is their own model or one provided by a third party.⁴⁷ However, this definition is not entirely comprehensive as, in practice, several scenarios may emerge from the activities of downstream providers modifying and/or integrating GPAI models within systems. In certain cases, they may be regarded as providers of the GPAI model itself (3.1.); in others, they may qualify as providers of a GPAI system (3.2.); and in yet other circumstances, they may fall within the category of high-risk AI system providers (3.3.).

3.1. Downstreaming into (new) GPAI Models

The AI Act’s narrow definition of downstream providers leaves a regulatory gap, as it fails to account for activities such as adapting or optimizing models, which are often referred to as downstream development within technical communities. This stage of the GPAI value chain is crucial, as it covers the various ways in which a GPAI model can be modified through fine-tuning or other adjustments for a specific application, function, or domain (for example, healthcare), or adapted more broadly into a different general-purpose model.⁴⁸ Such activities play a central role in the GPAI market. Notable examples include the creation of Deepseek LLM in 2024, based on the architecture of Meta’s LLaMA,⁴⁹ and Facebook’s development of

⁴⁷ AI Act, art 3(68).

⁴⁸ *ibid.*

⁴⁹ DeepSeek-AI and others, ‘DeepSeek LLM: Scaling Open-Source Language Models with Longtermism’ (arXiv, 5 January 2024) <<http://arxiv.org/abs/2401.02954>> accessed 12 May 2025.

RoBERTa in 2019 by modifying Google's BERT.⁵⁰ However, it is unclear whether or how a modifier may fall under the definition of a provider in Article 3(3) of the AI Act.

Recital 97 of the AI Act recognises that AI models, including GPAI models, can be 'modified or fine-tuned' to create new models tailored to particular tasks or domains. However, this practice introduces significant legal uncertainties under the AI Act. While Article 3(23) of the Regulation defines substantial modification in the context of AI systems, it does not provide a corresponding definition for modifying or fine-tuning GPAI models. As a result, it is unclear at what point such modifications constitute new models that would trigger provider-level obligations under Chapter V for the downstream modifier.

The legal implications of this element of the value chain are further complicated by the variety of technical methods involved. As academic commentary has highlighted, different techniques may have markedly different regulatory consequences. For instance, Reinforcement Learning from Human Feedback (RLHF) and Low-Rank Adaptation (LoRA) entail modifications to the model's internal parameters or architecture, which could result in the creation of a new GPAI model. In contrast, Retrieval-Augmented Generation (RAG) typically involves connecting a model to external databases or tools without altering its core architecture, which creates uncertainty as to whether a new model has been created.⁵¹ Additional complexity arises with respect to GPAI models with systemic risks. Fine-tuning can alter a model's capabilities, potential uses, or overall behaviour, thereby modifying its risk profile and potentially triggering new obligations under Article 55 and Chapter V.

In a recent contribution, legal scholars argued that there are two possible interpretations of the obligations of those who modify a GPAI model.⁵² On the one hand, the AI Act could be interpreted as not covering this scenario at all. This view is supported by the wording of Article 25, which focuses exclusively on modifications to AI systems including GPAI systems, without mentioning GPAI models. According to this interpretation, an entity that fine-tunes a GPAI model would not be considered a provider of that model—even if the fine-tuning constitutes a substantial modification—since the provision does not extend provider obligations to modifications of models.

On the other hand, these scholars argue that certain forms of modification by deployers may indeed give rise to provider responsibilities. Specifically, if such actions lead to a substantial modification or result in the creation of an entirely new model, the modifier may be considered a new model provider under the AI Act. This interpretation is supported by Recital 109, which states that, in the 'case of a modification or fine-tuning of a model, the obligations for providers of general purpose AI models should be limited to that modification or fine-tuning, for example by complementing the already existing technical documentation with information on the modifications, including new training data sources, as a means to comply with the value chain obligations provided in this Regulation'. According to these scholars, Recital 109 acknowledges that the value chain obligations under Article 25 are limited and that obligations

⁵⁰ 'RoBERTa: An Optimized Method for Pretraining Self-Supervised NLP Systems' (*MetaAI*, 29 July 2020) <<https://ai.meta.com/blog/roberta-an-optimized-method-for-pretraining-self-supervised-nlp-systems/>> accessed 20 July 2024.

⁵¹ Benjamin Blum and Jakob Rappenglück, 'Fine-Tuning von GPAI-Modellen nach der KI-Verordnung: Eine Regelungslücke für Zukunftstechnologie?' (2024) 40 *Computer und Recht* 626.

⁵² Philipp Hacker and Matthias Holweg, 'The Regulation of Fine-Tuning: Federated Compliance for Modified General-Purpose AI Models' (Social Science Research Network, 11 June 2025) <<https://papers.ssrn.com/abstract=5289125>> accessed 16 June 2025.

should apply to modifiers in certain circumstances. The underlying rationale is that confining obligations solely to the original provider could result in regulatory gaps, particularly when modifications introduce novel risks that were neither anticipated nor managed by the initial developer. Furthermore, these scholars argue that, by analogy and by applying the definition of Article 3(23), a modification to a GPAI model would be deemed as 'substantial' if it altered its risk profile. Accordingly, as a precautionary approach, entities engaging in significant modifications of GPAI models should consider the possibility that they may acquire provider status and, consequently, become subject to the corresponding obligations under the AI Act.

A different approach has been recently proposed by Pacchiardi and others.⁵³ These authors suggest that the modification should be measured in relation to changes to the model's behavioural characteristics. This would be done by directly measuring the difference in behaviour, or by establishing a set of proxy criteria (e.g. monetary cost of the modification, amount of compute, amount and variety of data used, and strength of the alteration applied) and conduct a set of experiments to determine differences in capability profiles between original and altered models.

The non-binding guidelines on the scope of obligations for providers of GPAI models under the AI Act have recently clarified the Commission's position with regard to future enforcement. While the Commission acknowledges that the final interpretation will be in the hands of the Court of Justice of the European Union (CJEU),⁵⁴ its position is that only modifications leading to a 'significant' change to the 'model's generality, capabilities, or systemic risk' will trigger provider obligations.⁵⁵ Notably, the Commission has introduced the term 'significant change' instead of 'substantial modification', although this difference in wording does not appear to reflect a difference in approach.

The Commission also establishes that a criterion for indicating such a change is that the computing power required for the modification is greater than one third of that used for the original model. This is based on the idea that such an amount of computing power indicates a significant change.⁵⁶ Furthermore, if a GPAI model with systemic risk is modified and the computing power meets the threshold, the new model is also presumed to have systemic risk.⁵⁷ It is worth noting that academic commentary has suggested that the Commission's intention is to minimise the likelihood of downstream modifiers falling within the GPAI model provider category, given that the threshold is set so high that it leaves most modified models outside its scope. Likewise, as the threshold for qualifying as a GPAI model with systemic risk is already high, it is even more difficult for modifiers to be considered as providers of this category.⁵⁸

If the CJEU were to take a different approach than the one given by the Commission's guidelines, namely considering that downstream modifiers do not qualify as providers, the

⁵³ Lorenzo Pacchiardi and others, 'A Framework to Categorise Modified General-Purpose AI Models as New Models Based on Behavioural Changes: Collection of External Scientific Studies on General Purpose AI Models under the EU AI Act.' (European Commission Joint Research Centre 2025) <<https://data.europa.eu/doi/10.2760/4372557>> accessed 17 October 2025.

⁵⁴ Commission (n 17), para 9.

⁵⁵ *ibid*, para 62.

⁵⁶ *ibid*, para 63.

⁵⁷ *ibid*, para 70.

⁵⁸ Lilian Edwards, Commentary on Article 51, in: Ebers/Zou (eds.), *Artificial Intelligence Act – A Commentary* (Beck/Hart/Nomos, forthcoming).

legal framework that would apply to them is the GPSR. The scope of that regulation is further explained in Section 5.1. of this article.

3.2. Downstreaming into GPAI systems

It is clear from the wording in Articles 3(66) and 3(68), as well as from Recital 100 of the AI Act, that incorporating a model into a system triggers obligations related to AI system providers. In such cases, the criteria set out in Article 6, in conjunction with Annexes I and III, must be applied to determine whether the resulting system qualifies as high-risk. This triggers additional compliance requirements, depending on the system's intended purpose and application domain.

GPAI models can be integrated into different AI systems, including among those 'general-purpose AI systems'.⁵⁹ These systems have the capacity to serve 'a variety of purposes, both for direct use as well as for integration in other AI systems'.⁶⁰ This wording makes clear that GPAI systems are, by design, not tied to a specific application or domain. Consequently, they do not fall under the high-risk category given by Article 6 of the AI Act, which depends on a system's specific intended purpose and context of use. Since GPAI systems are intended for general applicability across multiple use cases, they do not meet the threshold for categorisation within a particular risk area as outlined in Annex I or Annex III of the AI Act.⁶¹ This interpretation is further supported by Article 25(1)(c), which clarifies that the obligations for providers of high-risk AI systems under Article 16 only apply for those who modify the intended purpose of a GPAI system in such a way that it becomes a high-risk AI system under Article 6.

As GPAI systems do not fall within the high-risk category, providers of these technologies are only subject to basic horizontal obligations. These include taking measures to ensure their staff and other persons involved in operating the system have a sufficient level of AI literacy.⁶² In addition to this, Article 50 of the AI Act imposes certain transparency obligations on these providers, some of which relate to prompt moderation strategies, as discussed in subsection 1.2 of this article, as strategies to prevent undesired outputs from particular user prompts.

In that sense, Article 50(1) requires providers of systems intended for interaction with human users, such as chatbots, to disclose that they are dealing with an AI system. Alternatively, it must be obvious to a 'reasonably well-informed, observant and circumspect' user that they are interacting with an AI system. This can be achieved by adding disclosures to the interface or by including such information in the system's output.

Furthermore, providers of AI systems that generate synthetic audio, image, video or text content must ensure that the output can be identified as artificially generated or manipulated. This obligation extends beyond introducing 'watermarks' that could be removed by the end user, instead requiring the implementation of technical solutions that are 'effective, interoperable, robust and reliable as far as this is technically feasible', taking into account in

⁵⁹ AI Act, art 3(63) .

⁶⁰ AI Act, art 3(66).

⁶¹ Martin Ebers, 'Regulierung generativer Künstlicher Intelligenz in der KI-VO', in: Martin Ebers/Benedikt Quarch (eds.), *Rechtshandbuch ChatGPT: KI-basierte Sprachmodelle in der Praxis* (Nomos Verlagsgesellschaft 2024), pp. 43 et seq. at p. 80; Fernández-Llorca and others (n 34).

⁶² AI Act, Art 4.

light of the type of content, implementation costs and the state of the art of these solutions.⁶³ In this context, moderating text output would not appear to serve a practical purpose at the first glance, as users can copy and paste, or transcribe, the text for their intended use. However as shown in subsection 7.1.2 below, output moderation can still play a key role in the deployment of services in social media scenarios. In the case of images, videos or audio, providers could, for instance, incorporate fingerprints that are not necessarily visible, but which can be detected by automated tools implemented by social media platforms for their own moderation processes.⁶⁴

It can be concluded at this point that the distinction between GPAI models and GPAI systems creates a regulatory blind spot, which is problematic. While systemic risk is recognised at the model level, it is not adequately addressed at the systems level. This results in a paradox whereby systems that pose systemic risks, such as many publicly available chatbots or AI companions, may evade meaningful oversight, unless they happen to fall within predefined high-risk categories.⁶⁵

3.3. Downstreaming into high-risk AI systems

As such, GPAI downstream providers qualify as high-risk providers in two scenarios: First, when they incorporate a GPAI model, regardless of whether it poses a systemic risk, into a system that falls within the high-risk category; second, pursuant to Article 25(1)(b) and 25(1)(c), when they modify a GPAI system's intended use in such a way that it becomes a high-risk system. In these cases, Article 25(2) first and second sentence establish that the original provider retains its obligations with respect to the original system but is not regarded as the provider of the modified system. Instead, the initial provider is subject to a set of cooperation and information-sharing obligations towards the new provider, as detailed in the following subsection. However, Article 25(2) third sentence states that the initial providers can avoid these responsibilities by clearly specifying that its AI system is not to be changed into a high-risk AI system.

Article 25(4) covers scenarios in which providers of high-risk AI systems use or integrate the systems, tools, services, components or processes of third parties. In such cases, both the provider and the third party must reach agreements on how to share any necessary information, capabilities, technical access or assistance to help the provider comply with the obligations set out in the AI Act.

Such cooperation and information sharing are crucial, as being classified as high-risk triggers a wide range of regulatory requirements under the AI Act, most notably the obligation to undergo a conformity assessment procedure to demonstrate compliance with the more specific requirements for such systems.⁶⁶ This includes adopting risk management systems⁶⁷ and data governance practices,⁶⁸ providing heightened technical documentation⁶⁹ and record-

⁶³ AI Act, art 50(2).

⁶⁴ Felipe Romero Moreno, "Upload Filters" and Human Rights: Implementing Article 17 of the Directive on Copyright in the Digital Single Market' (2020) 34 International Review of Law, Computers & Technology 153.

⁶⁵ Samuel Carey, Regulating Uncertainty: Governing General-Purpose AI Models and Systemic Risks, (2025) EJRR 1-17.

⁶⁶ AI Act, art 16(f).

⁶⁷ *ibid*, art 9.

⁶⁸ *ibid*, art 10.

⁶⁹ *ibid*, art 11.

keeping,⁷⁰ using appropriate benchmarks to achieve accuracy, robustness, and cybersecurity,⁷¹ as well as drafting instructions and adopting other relevant measures to ensure sufficient transparency of the system's operation⁷² and human oversight by the end user.⁷³ They must also take corrective measures if their systems are found not to be in conformity with the obligations of the AI Act and inform relevant stakeholders, such as distributors and deployers.⁷⁴ Additionally, they must provide the relevant authorities with information about their compliance with the Regulation, and designate an authorised representative within the EU.⁷⁵

3.4. Prohibited AI practices

Article 5 of the AI Act establishes a list of AI practices that are forbidden from being placed on the market or put into service or use. These include practices involving the use of subliminal techniques with the objective or the effect of materially distorting the behaviour of individuals or groups in a way that causes, or could reasonably be expected to cause, significant harm to them or to third parties. Similarly prohibited are systems that exploit the vulnerabilities of individuals with the objective or effect of such behavioural distortion.

In February 2025, the Commission published guidelines on the prohibited practices under the AI Act. The guidelines state that these prohibitions apply to both GPAI systems and systems with 'intended purposes'. According to the guidelines, deployers are expected not to engage in prohibited practices and providers must refrain from placing into the market or put into service systems that are reasonably likely to behave or be directly used in a prohibited manner. With regard to GPAI systems, the guidelines provide examples and state that providers must take 'appropriate and proportionate measures' to prevent such uses and ensure that no harm is caused.⁷⁶

Nevertheless, there are several practical challenges at this stage. One is that, given their position in the value chain, deployers will have very little information with which to judge how likely it is that prohibited practices will emerge. The second relates to the benchmarks against which the 'reasonably' likelihood of a prohibited use should be measured in order to enable providers to take the appropriate measures or even refrain from releasing the product.

4. Cooperation duties along the AI value chain under the AI Act

In practice, cooperation between upstream and downstream providers involves a two-way exchange. Upstream providers hold essential information about a model's characteristics and how it functions. This information is critical for enabling system integration and for meeting the

⁷⁰ *ibid*, art 12.

⁷¹ *ibid*, art 15.

⁷² *ibid*, art 13.

⁷³ *ibid*, art 14.

⁷⁴ *ibid*, art 20.

⁷⁵ *ibid*, art 21.

⁷⁶ Commission, 'Commission Guidelines on prohibited artificial intelligence practices established by Regulation (EU) 2024/1689 (AI Act)' C(2025) 5052 final.

legal obligations imposed on system providers. In turn, downstream providers generate information on how the model interacts with other technologies and end users. This information is key to understanding how systemic risks may materialise. Furthermore, while upstream providers can introduce guardrails in their models to meet regulatory requirements, downstream providers may either introduce additional safeguards or implement modifications that override or undermine those introduced at the upstream level.⁷⁷

4.1. Upstream to downstream

The AI Act imposes several obligations on GPAI model providers to mitigate the operational risks posed by downstream providers and facilitate their integration into systems.⁷⁸ These duties are primarily set out in Article 53(1)(b) of the AI Act, which states that upstream providers must make information and documentation available to downstream providers that allows them to understand the ‘capabilities and limitations of the general-purpose AI model’, and to comply with their obligations under the AI Act, including descriptive information and technical elements for its integration.⁷⁹ While this information may be relevant for fine-tuning a model or incorporating it within a system, it does not imply access to the training data or to its source code. Notably, the information required by Article 53(1)(b) for the integration of models is not intended to affect the protection of ‘intellectual property rights and confidential business information or trade secrets’, such as the model’s source code.

As explained in Section 2.2 of this article, the information required under Article 53(1)(b) is detailed further in the Transparency Chapter of the Code of Practice and is collected through the Model Documentation form it introduces. The ‘Use’ section of the form is particularly relevant for the cooperation with downstream providers, as it outlines the model’s acceptable use policy, its intended purpose, and the types of AI systems into which it can be integrated.⁸⁰ Furthermore, the Code of Practice clarifies that, without prejudice to intellectual property, confidentiality and trade secret protections, signatories must provide additional information to downstream providers to enable them to properly understand of the capabilities and limitations of the general-purpose AI model, both for its integration into their own AI systems and to ensure compliance with their obligations under the AI Act.⁸¹

The Code acknowledges that the information required under this Article of the AI Act may not be sufficient to meet legal and practical needs on its own. It therefore stipulates that, where necessary to understand the model’s capabilities and limitations during integration and to enable compliance with AI Act obligations, upstream providers must supply additional information upon request from downstream providers.⁸² It should be noted, however, that the wording of the Code of Practice is unlikely to cover all the regulatory requirements that downstream providers must adhere to beyond the AI Act, including the GPSR and the GDPR, to name but a few.

⁷⁷ Sophie Williams, Jonas Schuett and Markus Anderljung, ‘On Regulating Downstream AI Developers’ [2025] *European Journal of Risk Regulation* 1.

⁷⁸ *ibid.*

⁷⁹ AI Act, art 53(1)(b).

⁸⁰ Oliver and Bommasani (n 41).

⁸¹ *ibid.*

⁸² *ibid.*, Measure 1.2.

In addition, Article 25(4) stipulates that when integrating ‘an AI system, tools, services, components, or processes’ from a third party into a high-risk AI system, the provider and the third party must regulate their cooperation via a contract. They must specify the information, capabilities, technical access, and other assistance required to enable the high-risk AI system provider to comply with the Regulation in a written agreement. While GPAI Models are not explicitly mentioned in this provision, they can be considered either ‘tools’ or ‘components’. Furthermore, this is also visible due to the final sentence of this paragraph, which stipulates that this provision does not apply to open-source tools, services, processes, or components that are not GPAI models.

Reading Article 53(1)(b) in conjunction with Article 25(4) suggests that, if a GPAI model is being integrated into a high-risk AI system, the GPAI provider must include the necessary information in a written agreement to enable the system provider to fulfil the obligations laid down in Article 16 of the AI Act. Therefore, GPAI providers should provide the necessary information to enable the system provider to: (a) ensure that the system complies with core requirements relating to risk management, data quality and governance, technical documentation, record-keeping, transparency, human oversight, and accuracy, robustness and cybersecurity;⁸³ (b) implement a quality management system;⁸⁴ (c) retain the logs automatically generated by the model to the extent they fall under the system provider’s control;⁸⁵ (d) carry out a conformity assessment prior to placing the system on the market or putting it into service;⁸⁶ (e) draw up the EU declaration of conformity in accordance with Article 47;⁸⁷ (f) take corrective action and provide relevant information to market surveillance authorities when there is evidence that the system may not be in conformity;⁸⁸ (g) demonstrate conformity with the applicable requirements upon a reasoned request by a competent authority;⁸⁹ and (h) ensure that the system meets relevant EU accessibility requirements.⁹⁰

The Safety and Security Chapter of the Code of Practice sets out a series of principles relevant to cooperation from upstream to downstream. As such, Principle A enshrines the principle of Appropriate Lifecycle Management, stating that the continuous assessment and mitigation of systemic risks throughout the entire model lifecycle, including after deployment, should involve cooperation with all the actors in the value chain. Similarly, Principle e establishes a principle of cooperation, emphasising the importance of collaboration with downstream modifiers and providers is important in systemic risk assessment and mitigation. This is reinforced by Principle B on Contextual Risk Assessment and Mitigation, which states that these assessments should include ‘as reasonably foreseeable’ system architectures and other software into which the model will be integrated. This clearly requires cooperation with the downstream providers.

However, except for some specific provisions, the duties of cooperation are not fleshed out in detail. Only Measure 9.1, which relates to serious incident reporting, explicitly refers to downstream modifiers or providers. It requires providers to enable these parties to report relevant information about such incidents to the respective provider, the AI Office, and national

⁸³ AI Act, art 16(a).

⁸⁴ *ibid*, art 16(c).

⁸⁵ *ibid*, art 16(e).

⁸⁶ *ibid*, art 16(g).

⁸⁷ *ibid*, art 16(j).

⁸⁸ *ibid*.

⁸⁹ *ibid*.

⁹⁰ *ibid*.

authorities. In addition to this, measure 5.1. on safety mitigation establishes that providers should offer ‘tools for other actors to use to mitigate the systemic risks stemming from the model’.

Finally, it is worth noting that, while Article 25(4) provides for contractual cooperation between upstream and downstream providers, the AI Act does not address whether liability can be shifted or excluded by contract.⁹¹

4.2. Downstream to upstream

As explained, downstream providers are well placed to gather information relevant to understanding how risks may materialise in practice. This is particularly relevant in the relationship between downstream providers and upstream providers of GPAI models with systemic risk, given the focus of the AI Act to mitigate ‘foreseeable’ risks.⁹² While this concept creates ambiguity regarding the level of duty of care that providers should apply, it also signals the need for extensive information gathering.⁹³ Arguably, the more information is gathered and shared across streams through adequate cooperation, the greater the extent to which risks become foreseeable.

In that sense, the obligation set by Article 53(1)(a) and substantiated by the Transparency Chapter for model providers to keep ‘up-to-date’ technical documentation of the model would benefit from cooperation at all levels, from downstream to upstream. However, the Model Documentation form introduced in that Chapter makes no reference to this cooperation or to information collected after model integration or system deployment.

Arguably, the obligation to implement a copyright policy discussed in Section 2.2. of this article may involve prompt moderation to prevent copyright infringement in outputs and, as Quintais notes, it must also ensure the protection of legitimate uses of copyrighted content, such as parody, caricature, or pastiche.⁹⁴ Although the majority of responsibility lies with the model provider, the downstream provider also plays a pivotal role due to their direct relationship with the end user. At a minimum, they should refrain from undermining the protective measures implemented by the model provider and report any issues identified during the operation of the system to them. However, the Copyright Chapter of the Code of Practice does not acknowledge this, as most references to the upstream–downstream relationship are limited to restrictions and safeguards introduced by the upstream provider, whether technical or set out in the acceptable use policies or similar documents.

Finally, the obligations relating to GPAI models with systemic risk, which focus primarily on the safety of the model, may require collaboration between the two levels of providers. This is

⁹¹ This was considered during the negotiations of New York’s upcoming RAISE Act; N.Y. Legis. Assemb. A6453A Reg. Sess. 2025-2026 <https://www.nysenate.gov/legislation/bills/2025/A6453/amendment/A> (RAISE act); Cf. Article 44-B, section 1423(2) of this proposal: “A provision within a contract or agreement that seeks to waive, preclude, or burden the enforcement of a liability arising from a violation of this article, or to shift that liability to any person or entity in exchange for their use or access of, or right to use or access, a large developer’s products or services, including by means of a contract of adhesion, is void as a matter of public policy.” However, the revised version of the bill no longer contains such a provision; <https://www.nysenate.gov/legislation/bills/2025/S6953/amendment/B>.

⁹² AI Act., art 3(65).

⁹³ Sheila Leunig and others, ‘The Colorado AI Act: A Compliance Handshake Between Developers and Deployers’ (The Policy Update 2025) <<https://mailchi.mp/policyupdate/coloradoaiact>> accessed 8 September 2025.

⁹⁴ João Pedro Quintais, ‘Generative AI, Copyright and the AI Act’ (2025) 56 Computer Law & Security Review 106107.

evident in the Safety and Security Chapter of the GPAI Code of Practice, which highlights the role of downstream providers in systemic risk assessment and mitigation. However, this chapter only stipulates that model providers should give them access to mitigation tools and facilitate the reporting of serious incidents by them.⁹⁵ It could be argued that, under such measures, mitigation tools should also encompass the establishment of cooperation channels for reporting information relevant to compliance that does not meet the threshold of a serious incident.

As has been emphasised, downstream providers can play a pivotal role in identifying, analysing and assessing risks. This is particularly relevant for post-market monitoring, given their ability to provide information on how the model is used in practice within a system. Some of the post-market monitoring measures outlined in Measure 3.5 of this Chapter require such cooperation, particularly the collection of information on breaches of the model's use restrictions, end-user feedback collection, frequent dialogue with affected stakeholders and community-driven model evaluation creation. Similarly, the examples of safety mitigations provided in Measure 5.1. include mechanisms that may require collaboration at this level, such as monitoring inputs and/or outputs, and mechanisms that facilitate cooperation by offering tools for risk mitigation.

4.3. Cooperation with deployers

The cooperation duties between providers and deployers along the value chain are limited mostly to the relationship between providers of high-risk AI systems and deployers. The burden of the requirements is primarily placed on the system provider. These include the obligation under Article 13 for providers to design and develop their systems in a transparent manner that enables interpretation by deployers. Furthermore, high-risk AI system providers must ensure that their systems are accompanied by concise, complete, correct and clear instructions for use that are accessible and comprehensible to deployers. Furthermore, Article 14 requires providers to design their systems in such a way as to allow meaningful human oversight during use.

Conversely, under Article 26(5), deployers are obliged to monitor the operation of the system in accordance with the instructions given for its use and to inform providers via their post-market systems of any relevant data required to evaluate ongoing compliance of the system obligations set out in the AI Act. Similarly, this article also requires deployers to inform providers, distributors and authorities in cases of emerging risks and of serious incidents.

Finally, it should be noted that Article 27 of the AI Act requires deployers to conduct fundamental rights impact assessments prior to deployment. However, this article does not impose any specific obligation on providers to provide information relevant to such an assessment.

4.4. Interim conclusion

Overall, while the framework set out by the AI Act and the Code of Practice acknowledges the need for cooperation between upstream and downstream providers, it does so in a fragmented

⁹⁵ Matthias Samwald and others (n 45).

and often implicit manner. Upstream providers are primarily responsible for supplying information and introducing safeguards, while the role of downstream providers is to generate insights on real-world risks. However, aside from specific provisions on incident reporting and access to mitigation tools, the existing framework falls short of establishing robust and continuous cooperation channels. This creates a discrepancy between the formal allocation of duties and the practical requirements for ensuring effective risk management across the value chain. In this respect, despite acknowledging cooperation as a relevant principle, it is largely confined to contractual discretion, which may be hindered by asymmetries arising from the bargaining power of large model providers. In light of this, future developments should include standardised disclosure protocols.⁹⁶ Finally, although system providers have some specific cooperation duties with deployers, these are limited to the case of high-risk systems.

5. Obligations of providers under EU product safety and liability law

The AI Act is not the only piece of legislation applicable to AI at EU level. Rather than replacing previous legislation or pre-empting the creation of new laws, the Regulation clearly states that it is intended to be applied alongside existing EU legislation, including that on product safety⁹⁷ and on product liability.⁹⁸ This is important because the AI Act imposes strict requirements on high-risk systems, whereas GPAI models, due to their general nature, do not fall into this category. Recital 166 signals the EU legislature's position that even non-high-risk systems must be safe, with the GPSR acting as a 'safety net', complemented by liability rules under the Product Liability Directive. The following sections outline the obligations under these instruments.

5.1. General Product Safety Regulation (GPSR)

5.1.1. Scope and core obligations under the GPSR

The GPSR is designed to establish basic product safety rules in cases where no specific EU legislation exists, or where existing legislation does not cover certain aspects or risks.⁹⁹ It should be noted, however, that specific product safety legislation applies to the use of AI in certain fields and takes precedence over the GPSR, the Medical Device Regulation,¹⁰⁰ the Machinery Regulation¹⁰¹ and the Toy Safety Directive.¹⁰²

⁹⁶ Leunig and others (n 93).

⁹⁷ AI Act, art 2(9).

⁹⁸ AI Act, Recital 9; Directive (EU) 2024/2853 of the European Parliament and of the Council of 23 October 2024 on liability for defective products and repealing Council Directive 85/374/EEC 2024, Recital 13.

⁹⁹ *ibid*, arts 1(2) and 2(1).

¹⁰⁰ Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC [2017] OJ L 117, 5.5.2017, pp. 1–175.

¹⁰¹ Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC of the European Parliament and of the Council and Council Directive 73/361/EEC [2023] OJ L 165, 29.6.2023, pp. 1–102.

¹⁰² Directive 2009/48/EC of the European Parliament and of the Council of 18 June 2009 on the safety of toys [2009] OJ L 170, 30.6.2009, pp. 1–37.

The GPSR's definition of 'product' is somewhat limited in the context of AI, as it only refers to 'items' that may or may not be 'interconnected' with other items.¹⁰³ At first glance, this seems to refer only to physical products. Whether the GPSR applies on AI models and systems depends on whether the term 'item' is deemed to cover tangible or intangible items. With regard to 'software', it is only clear that this Regulation only applies to software that is embedded in a physical product.¹⁰⁴ However, recital 24 seems to suggest a broader interpretation, stating that 'Items which connect to other items or non-embedded items which influence the way another item works can present a risk for the safety of the product.'

An explanatory website on the Safety Gate portal, created under Article 27 of this Regulation claims that the legislation applies to 'products that are tangible, non-tangible or of a mixed nature, including apps and software products'.¹⁰⁵ Similarly, a Commission Q&A published in 2024 states that the aforementioned product definition applies to standalone software, though it is 'not to be seen as a software regulation as such'.¹⁰⁶ While these statements are non-binding, they may indicate which products will be subject to enforcement by the Commission under Chapter VI of the Regulation. The applicability of this Regulation in the AI context is further clarified by the criteria it establishes for assessing product safety. These include, inter alia, the appropriate cybersecurity features necessary to prevent cyberattacks and other external influences, as well as the 'evolving, learning and predictive functionalities of the product'.¹⁰⁷ Moreover, an examination of the legislative history of the GPSR reveals that the impact assessment accompanying its proposal identified one of the shortcomings of the previous legislation as its failure to cover software. The assessment also noted that the legislative proposal covered software.¹⁰⁸

The preceding points raise the possibility that software would fall under definition of a 'product' according to the GPSR. However, until the CJEU issues a judgment on this matter, it remains unclear whether stand-alone software and other intangible products now fall under the product definition according to the GPSR. One argument against this is that, unlike Article 4(1) of the 2024 Product Liability Directive (which will be explained in subsection 5.2.), Article 3(1) of the GPSR still does not explicitly include software in the product definition, despite this being discussed repeatedly during the legislative process.¹⁰⁹ Consequently, many authors assume that the GPSR does not cover stand-alone software.¹¹⁰ Other authors, in addition to the aforementioned Commission position, take the view that – analogously to product liability law

¹⁰³ GPSR, art 3(1).

¹⁰⁴ See for instance: GPSR, art 15(3)(a) and 15(3)(b).

¹⁰⁵ 'Safety Gate: The EU Rapid Alert System for Dangerous Non-Food Products' <<https://ec.europa.eu/safety-gate/#/screen/pages/productSafetyLegislation>> accessed 12 May 2025.

¹⁰⁶ EU Commission, 'EU General Product Safety Regulation Frequently Asked Questions (FAQ)' <<https://ec.europa.eu/safety-gate/#/screen/pages/obligationsForBusinesses>> accessed 7 August 2025.

¹⁰⁷ GPSR, arts 6(1)(g) and 6(1)(h).

¹⁰⁸ Commission 'Commission staff working document impact assessment Accompanying the document Proposal for a Regulation of the European Parliament and of the Council on general product safety, amending Regulation (EU) No 1025/2012 of the European Parliament and of the Council, and repealing Council Directive 87/357/EEC and Directive 2001/95/EC of the European Parliament and of the Council' SWD(2021) 168 final.

¹⁰⁹ see Sebastian Rockstroh, EU-Cybersicherheitsrecht für Produkte, DuD 2023, 332 ff., 335; EU Commission, Report on the safety and liability implications of Artificial Intelligence, the Internet of Things and robotics of February 19, 2020, 11.

¹¹⁰ Carsten Schucht and Gerhard Wiebe, *Eu-Produktsicherheitsverordnung: General Product Safety Regulation: Gpsr* (Nomos 2024); Sebastian Rockstroh, 'EU-Cybersicherheitsrecht für Produkte: Ein Überblick à Der Flickenteppich braucht Struktur' (2023) 47 Datenschutz und Datensicherheit - DuD 332; Nils Löfing, 'Part 1: Product Compliance for Consumer Products Powered by AI' (*Bird & Bird*, 7 February 2024) <<https://www.twobirds.com/en/insights/2024/global/product-compliance-for-consumer-products-powered-by-ai>> accessed 15 September 2025.

– the term ‘product’ in Article 3(1) GPSR also covers purely digital products such as software.¹¹¹

The GPSR imposes principal obligations on ‘manufacturers’, which are defined as any natural or legal person who manufactures a product, or has it designed or manufactured and then markets it under their own name or trademark.¹¹² In addition, the Regulation establishes specific obligations for ‘providers of online marketplaces’, defined as those who use online interfaces (e.g. websites or apps¹¹³) to sell products.¹¹⁴ However, these obligations do not necessarily apply to software or AI, as they could be considered limited to online marketplaces for physical goods such as Amazon.

Arguably, both GPAI models and GPAI systems would qualify as ‘products’ under this Regulation, meaning that their providers would be considered ‘manufacturers’, triggering safety obligations. While there is some overlap between the obligations set out in this Regulation and those set out in the AI Act for those developing GPAI models, regardless of whether they pose systemic risks, this Regulation is also likely to address areas not covered by the AI Act. Furthermore, as both Regulations are enacted under the New Legislative Framework, the general nature of the GPSR would turn it into a safety net for AI applications. For GPAI systems, the GPSR would address the gaps in the AI Act, as these systems do not fall into any risk category and carry no specific safety obligations under the Act. In addition, the Regulation states that anyone who ‘substantially modifies’ a product will be considered its manufacturer for either the modified part or the entire product, provided that the change affects its safety. This would trigger safety obligations under the Regulation. In this respect, the GPSR could partly address the regulatory gap for downstream modifiers under the AI Act in cases where they do not qualify as new providers under the latter.

As explained in the introduction, the GPSR only applies to ‘consumer products’, where ‘consumer’ refers exclusively to natural persons acting in private contexts. However, products not primarily intended for consumers also fall within its scope if it is foreseeable that they could be used by consumers. In this way, the Regulation may indirectly affect commercial and professional activities.

Table 2 below shows the obligations set out in the GPSR and how these compare with the obligations for GPAI model and system providers under the AI Act. It should be noted, however, that while the GPSR provides generic requirements, the AI Act establishes more technical ones. Where obligations under the former overlap with those under the latter, the more specific obligations under the AI Act would take precedence.

The primary obligation imposed on manufacturers by the GPSR is to ensure that their products are ‘safe’.¹¹⁵ This obligation includes conducting risk assessments and implementing mitigation strategies and drawing up technical documentation describing the product and its safety-related characteristics.¹¹⁶ Furthermore, products must include instructions and safety

¹¹¹ Philipp Reusch, ‘KI Und Software Im Kontext von Produkthaftung Und Produktsicherheit’ 2023 *RD* 152; Clemens Mattheis, ‘Es Weihnachtet Schon Sehr, Da Steht Die Neue Produktsicherheitsverordnung Vor Der Tür!’ 2024 *CCZ* 325.

¹¹² *ibid.*, art 3(8).

¹¹³ *ibid.*, art 3(15).

¹¹⁴ *ibid.*, art 3(14).

¹¹⁵ *ibid.*, art 5.

¹¹⁶ *ibid.*, art 9(2).

guidelines, unless they can be used safely without them.¹¹⁷ If a product is deemed unsafe after its release, manufacturers must take corrective measures, which may include recalling the product, and must inform users and the relevant authorities.¹¹⁸ Under this Regulation, a 'safe product' is one that does not present any risk or only presents minimal risks that are compatible with its normal or reasonably foreseeable use.¹¹⁹ According to recital 23, the safety assessment should consider 'all relevant aspects of a product, in particular its characteristics, such as the physical, mechanical and chemical characteristics, and its presentation, as well as the specific needs and risks which the product represents for certain categories of consumers who are likely to use the products'.

Although the GPSR covers relevant issues such as physical and mental health, it does not address other problematic aspects of technologies such as AI, including discrimination, misinformation, and potential harm to democracy and fundamental rights.

Furthermore, this definition of safety can present challenges in the context of GPAI models and systems due to their generality and the wide range of activities for which they can be used, thereby broadening the scope of normal or reasonably foreseeable use. However, the Regulation provides GPAI model and system providers with a non-exhaustive list of criteria for assessing product safety, such as the potential effects of interaction with other products,¹²⁰ the products' cybersecurity features¹²¹ and its 'evolving, learning and predictive' capacities.¹²² The first factor involves assessing the potential impact of embedding a specific model within a particular interface, or of deploying GPAIs as agents that can interact with other software or websites.¹²³ The second factor could include a wide range of cybersecurity standards, including the capacity to respond to 'prompt hacking' techniques.¹²⁴ The third factor could encompass the impact of a model's evolving and learning capacities on existing safety measures.

There are other factors that are important for how safe a product is. These should apply to the whole AI value chain but would be especially important for downstream providers who adapt the GPAI model to a specific AI system. These factors include the characteristics of the target user group, such as identifying risks to vulnerable consumers including children, the elderly, and people with disabilities, and considering the gender-specific impacts on health and safety.¹²⁵ With regard to vulnerable consumers, a meaningful application of this could entail ensuring that the product meets the necessary digital literacy standards to guarantee safety. Moreover, this should also include addressing the risk of the GPAI system producing discriminatory or otherwise harmful results¹²⁶ that exploit user vulnerabilities.¹²⁷ With regard to

¹¹⁷ *ibid.*, art 9(7).

¹¹⁸ *ibid.*, art 9(8).

¹¹⁹ *ibid.*, art 3(2).

¹²⁰ *ibid.*, arts 6(1)(b) and 6(1)(c).

¹²¹ *ibid.*, art 6(1)(g).

¹²² *ibid.*, art 6(1)(h).

¹²³ Zhiheng Xi and others, 'The Rise and Potential of Large Language Model Based Agents: A Survey' (2025) 68 *Science China Information Sciences* 121101.

¹²⁴ Rababah and others (n 11).

¹²⁵ GPSR, art 6(1)(h).

¹²⁶ Prasanjit Rath and others, 'LLM Safety for Children' in Weizhu Chen and others (eds), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: Industry Track)* (Association for Computational Linguistics 2025) <<https://aclanthology.org/2025.naacl-industry.62/>> accessed 15 May 2025.

¹²⁷ For some approaches on the safety of children and AI that include considerations for different stakeholders, see: Nomisha Kurian, "'No, Alexa, No!': Designing Child-Safe AI and Protecting Children from the Risks of the "Empathy Gap" in Large Language Models' [2024] *Learning, Media and Technology* 1.

gender-differentiated impacts, safety assessments should also prevent outputs that constitute, or facilitate forms of gender-based violence as defined in the Directive on combating violence against women and domestic violence.¹²⁸ Examples include the generation of non-consensual sexual imagery, or content that is misogynistic or discriminatory. Regardless, the emphasis of the GPSR on physical safety may limit the possibility of an interpretation encompassing such elements.

Table 2: Specific safety obligations under the GPSR and their equivalence for GPAI models and systems in the AI Act			
Obligation for manufacturers under the GPSR	Equivalent obligation for GPAI model providers in the AI Act	Equivalent obligation for providers of GPAI models with systemic risk in the AI Act	Equivalent obligation for GPAI system providers in the AI Act
Conducting risk analysis and mitigation (Article 9(2))	[no equivalent obligation]	Article 55(1)(b)	[no equivalent obligation]
Drawing up technical documentation containing at least the product description and essential characteristics for assessing its safety (Article 9(2))	Article 53(1)(a) and Article 53(1)(b)	Article 53(1)(a) and Article 53(1)(b)	[no equivalent obligation]
Providing and disclosing a type, batch or serial number or other element enabling the identification (Article 9(5))	Article 53(1)(a); Annex XI, Section 1, 1(f)	Article 53(1)(a) Annex XI, Section 1, 1(f)	[no equivalent obligation]
Disclosing the name and contact information to the consumer (Article 9(6))	[no equivalent obligation]	[no equivalent obligation]	[no equivalent obligation]
Drafting instructions and safety information in an understandable	[no equivalent obligation]	Arguably covered by Article 55(1)(d)	[no equivalent obligation]

¹²⁸ Directive (EU) 2024/1385 of the European Parliament and of the Council of 14 May 2024 on combating violence against women and domestic violence [2024] OJ L, 2024/1385, 24.5.2024.

language (Article 9(7))			
If the manufacturer has reasons to believe the product is dangerous, take corrective measures, inform consumers and inform authorities (Article 9(8))	[no equivalent obligation]	Article 55(1)(c) (not including obligation to inform deployers/consumers)	[no equivalent obligation]
Inform other relevant stakeholders in the value chain about emerging safety concerns (Article 9(10))	[no equivalent obligation]	[no equivalent obligation]	[no equivalent obligation]
Disclosing relevant contact information for allowing consumers to file complaints and to submit information on safety concerns (Article 9(11))	[no equivalent obligation]	[no equivalent obligation]	[no equivalent obligation]
Respond to safety complaints and keep track of them (Article 9(12))	[no equivalent obligation]	Article 55(1)(c)	[no equivalent obligation]

5.1.2. Complementary role of the GPSR in relation to the AI Act

In summary, the GPSR addresses important regulatory gaps left by the AI Act in relation to GPAI, providing complementary safeguards.

First, with respect to GPAI models in general, the GPSR introduces baseline obligations absent from the AI Act, thereby raising the bar for safety and transparency. These include requirements to: conduct risk analysis and mitigation;¹²⁹ disclose manufacturer contact details;¹³⁰ provide safety instructions in accessible language;¹³¹ take corrective action and inform consumers and authorities when a product may be dangerous;¹³² notify other actors in the value chain about emerging risks;¹³³ establish internal complaint mechanisms for consumers;¹³⁴ and track and respond to such complaints.¹³⁵

¹²⁹ GPSR, art 9(2).

¹³⁰ *ibid*, art 9(6).

¹³¹ *ibid*, art 9(7).

¹³² *ibid*, art 9(8).

¹³³ *ibid*, art 9(10).

¹³⁴ *ibid*, art 9(11).

¹³⁵ *ibid*, art 9(12).

Second, for GPAI models with systemic risks, Article 55(1) establishes obligations relating to model evaluation, risk assessment and mitigation, documentation, reporting of serious incidents, and cybersecurity protection. However, the GPSR goes further. Whereas Article 55(1) of the AI Act limits reporting obligations to notifying the AI Office and national authorities of serious incidents, the GPSR requires manufacturers to inform both authorities and consumers whenever a product is believed to pose a danger. In addition, the GPSR obliges manufacturers to notify other actors in the value chain of safety concerns, while Article 55(1) does not. Moreover, the AI Act does not require the implementation of internal complaint-handling mechanisms or the provision of contact information to consumers (or deployers).

Third, regarding GPAI systems, the GPSR plays an essential role in light of the more limited provisions of the AI Act in this area. It can ensure that such systems are also subject to risk assessment, traceability, user guidance, and complaint handling obligations.

5.2. The new Product Liability Directive (PLD) 2024

5.2.1. The updated liability framework at EU level

The new Product Liability Directive (PLD 2024), adopted in October 2024 shortly after the AI Act, revises the 1985 Product Liability Directive (PLD 1985)¹³⁶ ‘in light of developments related to new technologies, including Artificial Intelligence’¹³⁷ in order to ensure that claimants “enjoy the same level of protection irrespective of the technology involved” when seeking damages.¹³⁸

A first novelty concerns the concept of ‘product’. Whereas Article 2 PLD 1985 referred only to movables and electricity, Article 4(1) PLD 2024 explicitly includes software. Recital (13) PLD 2024 clarifies that this encompasses operating systems, firmware, computer programs, applications and AI systems, and specifies that it is irrelevant whether the software is stored on a device, accessed via a communications network or cloud technologies, or provided through a software- as -a-service model.¹³⁹ Another major change is the extension of liability for harm caused by defective components. Under Article 8(1)(a) and (b) PLD 2024 manufacturers are liable for harm caused by defective components, defined broadly as ‘any item, whether tangible or intangible, raw material or related service, that is integrated into, or inter-connected with, a product.’

The PLD 2024 is primarily addressed to ‘manufacturers’, a term defined broadly enough to encompass both upstream and downstream providers. Under Article 4(5), a manufacturer is any person who ‘develops, manufactures or produces a product’, or who presents themselves as its manufacturer. Recital 13 further clarifies that this includes a ‘developer or producer of software, including AI system providers within the meaning of Regulation (EU) 2024/1689’.

¹³⁶ Council Directive 85/374/EEC of 25 July 1985 on the approximation of the laws, regulations and administrative provisions of the Member States concerning liability for defective products [1985] OJ 1985 L 210, 29.

¹³⁷ PLD, Recital (3), sentence 1.

¹³⁸ *ibid.*, Recital (3), sentence 4.

¹³⁹ *ibid.*, Recital (13).

Importantly, however, this reference does not suggest that GPAI model providers are excluded, nor does it make a clear distinction between upstream and downstream providers.

Moreover, Article 8(2) of the PLD 2024 broadens the definition of manufacturer to include anyone who ‘substantially modifies a product outside the manufacturer’s control’ and subsequently places it on the market or puts it into service. This provision is particularly relevant in the context of AI, where downstream providers often adapt or fine-tune GPAI models for specific applications. Whether a modification is ‘substantial’ should be primarily be assessed under the ‘relevant Union or national rules on product safety’; Article 4(18)(a) of the PLD 2024. Accordingly, AI systems must be assessed under Article 3(23) of the AI Act.

As the PLD 2024 also covers harm caused by defective components, liability may be shared between the manufacturer of a defective product and the manufacturer of a defective component integrated into that product.¹⁴⁰ The Directive explicitly requires Member States to ensure that liability in such cases may be imposed jointly and severally.¹⁴¹ Accordingly, GPAI model providers could be regarded as component manufacturers within the meaning of Article 4(4) of the PLD 2024, while downstream providers may be considered to be incorporating those components into a final product. Arguably, if a model is modified to such an extent that it does not constitute a new model, the original model would be considered a component of the final product. However, if the model modification is ‘substantial’ or ‘significant, the initial model would not be a component of the final product. In that case, it is the new model that would qualify as a component.

In the case of fine-tuning, if the pre-existing model constitutes a component of the fine-tuned model, the initial model provider would be liable for it as a component manufacturer. If the pre-existing model is considered to have been placed on the market, it is considered as a product covered by the PLD 2024; afterwards, when integrated into the final product, it is considered a component.

Liability under the PLD 2024 arises where ‘damage’ is caused by a ‘defective’ product. Under Article 9 of the PLD 1985, liability was limited to cases involving death, personal injury, and damage to property used for private purposes. The new Directive broadens this scope in two important ways. First, it explicitly recognises ‘medically recognised damage to psychological health’ as a form of personal injury.¹⁴² Second, it extends protection for the first time to the destruction or corruption of non-professional data, Article 6(1)(c) of the PLD 2024. However, several categories of harm commonly associated with AI systems are still excluded. These include violations of personality rights, discrimination, damage to commercially used items and pure economic loss.¹⁴³ Therefore, such harms fall outside the scope of the PLD 2024 and can only be pursued under national tort law.

The question of whether a product is defective is addressed in Article 7 of the PLD 2024. As in the PLD 1985, the benchmark remains the level of safety that the public can reasonably expect, considering the product’s presentation and its reasonably foreseeable use. The new Directive, however, refines and expands this concept. Notably, Article 7(1) of the PLD 2024

¹⁴⁰ *ibid*, arts 8(1)(a) and 8(1)(b).

¹⁴¹ *ibid*, art 12(1).

¹⁴² PLD, Recital (21).

¹⁴³ See PLD, Recital (24), first sentence: ‘Types of damage other than those provided for in this Directive, such as pure economic loss, privacy infringements or discrimination, should not by themselves trigger liability under this Directive.’

states that safety expectations must also be assessed against the safety standards ‘required under Union or national law’. By doing so, the PLD 2024 explicitly links EU product safety law with EU product liability law,¹⁴⁴ thereby reinforcing ‘the close relationship between product safety rules and liability rules’.¹⁴⁵ This has significant implications for AI systems: the safety requirements established under the AI Act and the GPSR must be regarded as minimum benchmarks when determining defectiveness.

Beyond this, the Directive provides a non-exhaustive list of additional factors for assessing defectiveness, many of which align with the GPSR. These include how the product interacts with other products;¹⁴⁶ its capacity for continuous learning or acquiring new features;¹⁴⁷ its compliance with cybersecurity requirements;¹⁴⁸ and whether it has been subject to recalls or other regulatory interventions.¹⁴⁹

5.2.2. Liability for GPAI under the PLD 2024

Whether manufacturers of GPAI models and/or systems can be held liable for damages caused by such models remains an open question.

The first difficulty arises from the scope of recoverable harm under the PLD 2024. Liability is limited to cases involving death, personal injury, damage to property, and corruption or destruction of data used for private purposes. Yet, GPAI outputs often result in other types of harm that are not covered by the Directive—such as violations of personality rights, privacy infringements, discrimination, or pure economic loss.

A second issue concerns the causal link between GPAI outputs and legally recognised forms of damage, and whether the new PLD 2024 covers wrongful information. GPAI typically generates outputs in the form of text, audio, or images. Such outputs do not directly cause death or personal injury, but they may contribute to it indirectly. For example, a user may commit suicide or engage in self-harm as a result of their interaction with a GPAI system¹⁵⁰ or might inflict harm on third parties based on erroneous or misleading information produced by it.¹⁵¹ Recital 13 of the PLD 2024 expressly excludes liability for ‘mere’ information, stating that ‘Information is not, however, to be considered a product, and product liability rules should therefore not apply to the content of digital files, such as media files or e-books or the mere source code of software.’ This exclusion could significantly limit the attribution of product liability to GPAI outputs themselves, not to mention the unresolved question of how to establish causation.

¹⁴⁴ In detail Ebers, *The New Product Liability Directive: Paving the Way for Follow-on Actions in Artificial Intelligence Law?*, in: Silvia Martinelli/Cristina Poncibo/Pierluigi Perri (eds.), *EU Product Liability: Platforms, Internet of Things, and Artificial Intelligence*, Routledge, 2025, pp. 153 et seq.

¹⁴⁵ PLD, Recital (46).

¹⁴⁶ *ibid.*, art 7(2)(d).

¹⁴⁷ *ibid.*, art 7(2)(c).

¹⁴⁸ *ibid.*, art 7(2)(f).

¹⁴⁹ *ibid.*, art 7(2)(g).

¹⁵⁰ ‘Parents of Teenager Who Took His Own Life Sue OpenAI’ (27 August 2025) <<https://www.bbc.com/news/articles/cgerwp7rdlvo>> accessed 1 September 2025.

¹⁵¹ Pat Tomlinson Chumney Richard, ‘ChatGPT Affirmed Greenwich Man’s Fears about His Mom before Murder-Suicide, YouTube Videos Show’ (*Greenwich Time*, 30 August 2025) <<https://www.greenwichtime.com/news/article/chatgpt-greenwich-ct-murder-stein-erik-soelberg-21022277.php>> accessed 1 September 2025.

The recital does not, however, rule out the possibility that software that generates harmful information could qualify as a defective product that leads to harmful results. Such a case is not comparable to the situation decided by the ECJ in the *Krone case*,¹⁵² which concerned incorrect health advice in a newspaper. While a newspaper that publishes incorrect information is merely a carrier of information, but does not produce it itself and therefore cannot be considered a defective product, the situation is different for software, at least if it is intended to produce specific outputs based on input data using a specially programmed algorithm.¹⁵³ In this case, software may well be classified as a defective product if it generates incorrect information contrary to legitimate safety expectations. Therefore, damages arising from the use of GPAI may include situations in which the interaction with a chatbot leads to suicide, self-harm, depression, or harm to third parties.¹⁵⁴

Arguably, the assessment of defectiveness in this context would differ depending on whether the product is a GPAI system such as ChatGPT, which is designed for broad use, or a companion AI that deploys a GPAI model within a specific application. This distinction is significant because it influences the evaluation of factors such as foreseeable use, interaction with other products, and the characteristics of the target user group.

Under the PLD 2024, upstream providers may be exempt from liability in three situations. First, where a GPAI model is integrated into a system outside the manufacturer's control (Article 8(1)(b)). According to Article 4(5), this applies if the manufacturer neither performed the integration, nor authorised or consented to it, and lacks the ability to provide software updates or upgrades. Second, exemption applies if the defectiveness of the GPAI system or model which is a component of a final product is attributable to that product's design or by the instructions given by the product manufacturer to the component manufacturer (Article 11(f)).¹⁵⁵ In addition to this, upstream providers may be exempt if they have a liability waiver agreement with the downstream provider.¹⁵⁶

Conversely, downstream providers may be exempt from liability if they substantially modify a product and the defect originates from a component unaffected by the modification.¹⁵⁷ Examples include cases where the defect originates from the original training data or the model's design, or where the model provider's safety measures are inadequate in light of the state of the art.

Finally, it should be noted that both upstream and downstream providers may be exempt from liability if the defect results from compliance with mandatory legal requirements¹⁵⁸ or if it was not possible to identify the defect based on the state of scientific and technical knowledge at the time the product was placed on the market (or during the period in which it remained under the manufacturer's control).¹⁵⁹

¹⁵² Case C-65/20 *VI v Krone Verlag Gesellschaft mbH & Co KG* EU:C:2021:471.

¹⁵³ See A Haftenberger, *Die Produkthaftung für künstlich intelligente Medizinprodukte. Eine Analyse der Anwendbarkeit und Eignung der europäischen Produkthaftungsrichtlinie 85/374/EWG* (Nomos 2023), 65 ff.

¹⁵⁴ Renwen Zhang and others, 'The Dark Side of AI Companionship: A Taxonomy of Harmful Algorithmic Behaviors in Human-AI Relationships' (arXiv, 2024) <<https://arxiv.org/abs/2410.20130>> accessed 20 May 2025.

¹⁵⁵ PLD, art 11(1)(f).

¹⁵⁶ *ibid*, art 12(2)(b).

¹⁵⁷ *ibid*, art 11(1)(g).

¹⁵⁸ *Ibid.*, art 11(1)(d).

¹⁵⁹ *ibid*, art 11(1)(e).

5.3. Interim results

Together, the GPSR and the PLD 2024 offer important additional layers of support to the AI Act when it comes to addressing responsibility and liability throughout the GPAI value chain. The GPSR closes some of the regulatory gaps by imposing baseline safety obligations. However, its contribution is limited in three respects: first, it focuses solely on product safety, ignoring broader issues such as the protection of fundamental rights; second, it lacks the technical specificity required for software, let alone GPAI; third, and most importantly, it remains unclear whether the GPSR applies to standalone software.

The PLD also plays a crucial complementary role, particularly because breaches of the AI Act or the GPSR could form the basis for establishing liability. This is evident from the wording of Article 7(1) of the PLD 2024, which states that non-compliance with safety requirements under EU law constitutes a cause of defectiveness. However, its scope is constrained in two ways. On the one hand, liability under the PLD is only triggered in cases of death, personal injury, or damage to property or data, excluding other harms frequently associated with GPAI such as violations of personality rights, discrimination, or reputational harm. On the other hand, it is unclear whether liability can be extended to GPAI-generated outputs, as Recital 13 expressly excludes ‘information’ from the notion of product. This ambiguity creates a potential loophole in the allocation of liability for harm caused by the wrongful or misleading outputs of GPAI systems.

6. Obligations for GPAI marketplaces under the DSA

The growing interest among developers in creating their own foundational models or adapting or optimising existing ones, has also given rise to ‘model marketplaces’. These are platforms where ‘users can upload AI systems and AI-related datasets, which in turn can be downloaded, and depending on the business model, queried, tweaked, or built upon by other users.’¹⁶⁰ Different platforms like Hugging Face, GitHub and Civitai have emerged as repositories with a vast amount of content consisting of models or relevant information for operating, creating or fine-tuning them, such as datasets.

These platforms act as intermediaries for hosting user-generated content, which may trigger responsibilities under both the DSA and the AI Act simultaneously. However, providing this kind of service raises questions and challenges regarding compliance with the existing EU regulatory framework. As Gorwa and Veale note, this framework may have limitations when interpreted in light of the characteristics of these types of platforms.¹⁶¹

GPAI marketplaces are likely to fall within the general definition of a ‘hosting’ intermediary and the specific category of an ‘online platform’ under the DSA. The former includes any service consisting of the storage of information provided by users at their own request¹⁶² and the latter

¹⁶⁰ Robert Gorwa and Michael Veale, ‘Moderating Model Marketplaces: Platform Governance Puzzles for AI Intermediaries’ (2024) 16 Law, Innovation and Technology 341.

¹⁶¹ *ibid.*

¹⁶² DSA, art 3(g)(iii).

comprises the storage and public dissemination of content at the users' request.¹⁶³ 'Information' and 'content' are used interchangeably in different parts of the DSA, and neither term is expressly defined. However, GPAI content (e.g. models, training data) uploaded to these marketplaces is very likely to be covered by them. This is evident from the Commission's interpretation in designating platforms such as Google Play and Apple's App Store as VLOPs, which offer a service that is relatively similar to GPAI marketplaces by first stating that they are hosting providers and platforms.¹⁶⁴ Furthermore, the General Court has recently ruled that the concept of 'information' used in the DSA must be understood 'in a broad sense'.¹⁶⁵

GPAI marketplaces are, however, very niche compared to other services, and are unlikely to reach the 45 million user threshold required for VLOP designation.¹⁶⁶ This would mean that they would have to comply with a less stringent set of requirements than Google and Apple. As Husovec explains, these obligations are 'meant to legislatively balance the liability assurances as to trust, safety and fairness' in the intermediary services.¹⁶⁷ A key component of this balance is to 'uphold *fair process* when moderating content'.¹⁶⁸ Moreover, the DSA sets out obligations relating to the handling of 'illegal content', which is defined as 'information that, in itself or in relation to an activity, including the sale of products or the provision of services, is not in compliance' with EU or Member States' law,¹⁶⁹ as well as permissions and guardrails for dealing with content that is legal but contrary to the platforms' terms of service.

Against this background, a key challenge for GPAI marketplaces is how broadly or narrowly 'content' is interpreted. This complexity and uncertainty surrounding the handling of apps by app stores¹⁷⁰ is exacerbated by the nature of GPAI models. Gorwa and Veale explain this by noting that models are not only content themselves, but are also capable of generating new content.¹⁷¹ The main challenge relating to this capacity of generate new content is spotting it, due to the complexity of the underlying neural networks.¹⁷² On the other hand, the question of content generated by the model relates to what should be targeted: is it a matter of the intended use of the GPAI? or of realised misuses? or of potential or likely uses?¹⁷³

These questions are critical for these marketplaces because, as hosting providers, they are exempt from liability for illegal content posted by their users, provided they do not have 'actual knowledge' of the illegal activity or content, nor awareness of 'facts or circumstances' that make the illegality apparent, and that they act promptly to remove or disable the content once they obtain such knowledge.¹⁷⁴

¹⁶³ DSA, art 3(i).

¹⁶⁴ Commission 'Decision designating Google Play as a very large online platform in accordance with Article 33(4) of Regulation (EU) 2022/2065 of the European Parliament and of the Council' C(2023) 2738 final, Recital 1; Commission 'Decision designating App Store as a very large online platform in accordance with Article 33(4) of Regulation (EU) 2022/2065 of the European Parliament and of the Council' C(2023) 2726 final, Recital 1.

¹⁶⁵ Case T-348/23 *Zalando SE v European Commission* EU:T:2025:821, para 29.

¹⁶⁶ Gorwa and Veale (n 160).

¹⁶⁷ Martin Husovec, *Principles of the Digital Services Act* (1st edn, Oxford University Press 2024) <<https://academic.oup.com/book/58088>> accessed 23 March 2025, p 22.

¹⁶⁸ *ibid*, p 23.

¹⁶⁹ DSA, art 3(h).

¹⁷⁰ Josh Cowls, Jessica Morley and Luciano Floridi, 'App Store Governance: Implications, Limitations, and Regulatory Responses' (2023) 47 Telecommunications Policy 102460.

¹⁷¹ Gorwa and Veale (n 160).

¹⁷² *ibid*.

¹⁷³ *ibid*.

¹⁷⁴ DSA, art 6(1).

Arguably, the interpretation given by the CJEU to such liability rules could help clarify how in-depth the examination of ‘content’ should be in the case of GPAI marketplaces.

6.1. No obligation to carry out model evaluations, audits or similar

When examining the equivalent provisions of the E-Commerce Directive that were replicated in the DSA, the CJEU limited the application of the aforementioned liability rules for illegal user-generated content by stating that ‘it is not sufficient that the provider was generally aware that its services are used to engage in some illegal activities’.¹⁷⁵ Instead, the provider must be notified of illegality ‘in a manner that is clear and precise so as to allow the provider to establish that infringement without conducting a detailed legal or factual examination’.¹⁷⁶ The CJEU reinforced this by stating that the illegality must ‘be specifically established or readily identifiable’.¹⁷⁷ Coupled with this is the fact that the DSA does not impose a general obligation on platforms to monitor content posted by their users.¹⁷⁸ At the same time, it allows platforms to carry out ‘voluntary own-initiative investigations’ in good faith and with diligence to address illegal content.¹⁷⁹

The first conclusion to be drawn from this is that providers of GPAI marketplaces would not be required to monitor the content uploaded by their users. In practice, this means that they would not be forced to test or audit the models uploaded by developers. In addition to this, a second conclusion would be that the marketplaces would be allowed to implement voluntary tests or audits on what is being uploaded.

6.2. Responding to user notices

Under the DSA, users can notify platforms of any alleged illegality in their services, which is considered as sufficient evidence of platforms knowing or being aware of it.¹⁸⁰ If read alongside the CJEU’s pre-DSA case law on actual knowledge, these notices should be ‘clear and precise so as to allow’ the marketplace to identify illegality ‘without conducting a detailed legal or factual examination’.¹⁸¹ Arguably, one notice would be more ‘clear and precise’ if it rightly signals that a model’s purpose is illegal (e.g. for producing child sexual abuse material). In addition, since ‘illegal content’ under the DSA includes content that does not comply with EU law, notices indicating that a model does not comply with safety legislation would also meet the criteria of clarity and precision. However, challenges may arise in the context of realised, potential or likely misuse. These may require substantiated quantitative and qualitative evidence, such as audits or tests depicting the percentage of harmful responses, or the severity and gravity of realised misuse (e.g. evidence that a GPAI model led to a single case

¹⁷⁵ Joined Cases C-682/18 and C-683/18 *Frank Peterson v Google LLC, YouTube LLC, YouTube Inc, Google Germany GmbH (C-682/18) and Elsevier Inc v Cyando AG (C-683/18)* EU:C:2021:503, para 35.

¹⁷⁶ *ibid*, para 35.

¹⁷⁷ *ibid*, para 113.

¹⁷⁸ DSA, art 8.

¹⁷⁹ *ibid*, art 7.

¹⁸⁰ *ibid*, art 16(3).

¹⁸¹ *Frank Peterson v Google LLC, YouTube LLC, YouTube Inc., Google Germany GmbH (C-682/18) and Elsevier Inc. v Cyando AG (C-683/18)* (n 175), para 35.

of suicide or self-harm would carry more weight than evidence of a single biased response to a prompt).

These notices activate a series of procedural safeguards under the DSA, for both the party issuing the notice and the user whose content is subject to a restriction affecting its availability, visibility, and accessibility. This includes the platform issuing a statement of reasons for their decision on the notice,¹⁸² as well as the possibility of filing an appeal with an internal appeals mechanism¹⁸³ and accessing different levels of redress, such as out-of-court dispute settlement,¹⁸⁴ or administrative¹⁸⁵ or judicial¹⁸⁶ redress. In the context of GPAI marketplaces, the requirement for statements of reasons to include ‘the facts and circumstances relied on in taking the decision’¹⁸⁷ can be understood as implying an obligation to disclose audits or other relevant evidence submitted by those who filed notices against the models. This information could be crucial for users who uploaded the models and want to file appeals or seek redress following a decision.

As mentioned previously, the existing marketplaces are far from reaching the 45 million user threshold required to classify them as VLOPs. However, it should not be dismissed that these or similar platforms could fall into that category in the future, particularly given how GPAIs lower the barrier to downstream development and use by individuals without technical training.¹⁸⁸ If this threshold is reached, these platforms would be required to conduct more rigorous due diligence processes, including risk assessment and mitigation strategies.

7. Obligations of VLOPs using GPAI

During the legislative process to adopt the AI Act, and in parallel with discussions on how to incorporate GPAI models within it, the co-legislators debated whether the new regulation should impose additional obligations on social media platforms that are classified as VLOPs under the DSA and that deploy recommender systems.¹⁸⁹ As this section shows, while the final version of the AI Act does not provide clear rules on this specific interaction between the DSA and the AI Act,¹⁹⁰ there are points of intersection that are becoming increasingly crucial from a compliance perspective.

The outcome of the legislative debate on the potential impact of the AI Act on social media platforms is reflected in Article 2(5) of the AI Act, which states that the new Regulation does not affect the liability rules for intermediary services under Chapter II of the DSA. Additionally, Recital 118 of the Act clarifies that, unless they pose ‘significant systemic risks’, AI models used by platforms follow the DSA’s risk-management framework.

¹⁸² DSA, art 17.

¹⁸³ *ibid*, art 20.

¹⁸⁴ *ibid*, art 21.

¹⁸⁵ *ibid*, art 53.

¹⁸⁶ *ibid*, art 21(1), art 50(3).

¹⁸⁷ *ibid*, art 17(3)(b).

¹⁸⁸ Tiago Lubiana and others, ‘Ten Quick Tips for Harnessing the Power of ChatGPT in Computational Biology’ (2023) 19 *PLOS Computational Biology* e1011319; Natalie Cooper and others, ‘Harnessing Large Language Models for Coding, Teaching and Inclusion to Empower Research in Ecology and Evolution’ (2024) 15 *Methods in Ecology and Evolution* 1757; Nishat Raihan, Christian Newman and Marcos Zampieri, ‘Code LLMs: A Taxonomy-Based Survey’, *2024 IEEE International Conference on Big Data (BigData)* (IEEE 2024) <<https://ieeexplore.ieee.org/document/10826108/>> accessed 23 May 2025.

¹⁸⁹ Kasia Söderlund and others, ‘Regulating High-Reach AI: On Transparency Directions in the Digital Services Act’ (2024) 13 *Internet Policy Review* <<https://policyreview.info/articles/analysis/regulating-high-reach-ai>> accessed 4 February 2025.

¹⁹⁰ Ugo Pagallo, ‘LLMs Meet the AI Act: Who’s the Sorcerer’s Apprentice?’ (2025) 1 *Cambridge Forum on AI: Law and Governance* e1.

In that sense, it should be noted that social media platforms like Facebook, X, or LinkedIn, who have been designated by the Commission as VLOPs, are among the different industries that are increasingly exploiting the benefits of GPAI in their activities. These platforms are incorporating GPAI models into various parts of their services, either as technical solutions for their moderation and recommendation processes, or as user services.

Hacker examined the interaction between the AI Act and the DSA, noting an overlap in requirements whereby VLOPs must address risks related to their platform activities, as well as those related to the implementation of AI systems.¹⁹¹ Consequently, platforms must, on the one hand, comply with rules designed to ensure ‘a safe, predictable and trusted online environment’¹⁹² for internet users, and on the other hand, comply with rules ensuring that their deployment of GPAI models within their services is ‘human-centric and trustworthy’ and protects ‘health, safety, fundamental rights enshrined in the Charter, including democracy, the rule of law and environmental protection’.¹⁹³

7.1. The two faces of GPAI use in social media platforms

GPAI has two faces when it comes to social media platforms: first, it is used internally to moderate¹⁹⁴ or recommend¹⁹⁵ content online. Second, and more visibly to the public, it is used by social media companies to develop new GPAI-based services. Examples of the latter include the release of the Grok chatbot on X, which can generate text and images, as well as reply to user queries when tagged in posts;¹⁹⁶ Meta’s deployment of AI-generated profiles;¹⁹⁷ Hinge’s new tool aimed at improving users’ dating profiles¹⁹⁸; and LinkedIn’s deployment of an AI post generator,¹⁹⁹ as well as the current testing of an advanced job search tool.²⁰⁰ News coverage of the latter company noted that its new terms of service include a discharge of liability for misleading content that may be generated by their users’ use of its GPAI-based

¹⁹¹ Philipp Hacker, ‘The AI Act between Digital and Sectoral Regulations’ <<https://www.bertelsmann-stiftung.de/doi/10.11586/2024188>> accessed 18 January 2025.

¹⁹² DSA, art 1(1).

¹⁹³ AI Act, art 1(1).

¹⁹⁴ Mohammad Nadeem and others, ‘Are Foundation Models the Next-Generation Social Media Content Moderators?’ (2024) 39 IEEE Intelligent Systems 70; Emmanuel Vargas Penagos, ‘ChatGPT, Can You Solve the Content Moderation Dilemma?’ (2024) 32 International Journal of Law and Information Technology eaae028.

¹⁹⁵ Dongxian Yu and others, ‘An AI-Driven Social Media Recommender System Leveraging Smartphone and IoT Data’ (2025) 81 The Journal of Supercomputing 272.

¹⁹⁶ ‘Elon Musk’s Startup Rolls out New Grok-3 Chatbot as AI Competition Intensifies’ *The Guardian* (18 February 2025) <<https://www.theguardian.com/technology/2025/feb/18/elon-musk-grok-3-ai-chatbot>> accessed 24 February 2025; ‘X Now Lets You Mention Grok in Replies, Get Explanations’ (*NewsBytes*, 3 August 2025) <<https://www.newsbytesapp.com/news/science/x-introduces-new-feature-to-interact-with-grok-ai-chatbot/story>> accessed 6 June 2025.

¹⁹⁷ Allison Morrow, ‘Meta Scrambles to Delete Its Own AI Accounts after Backlash Intensifies | CNN Business’ (*CNN*, 3 January 2025) <<https://www.cnn.com/2025/01/03/business/meta-ai-accounts-instagram-facebook/index.html>> accessed 24 February 2025.

¹⁹⁸ Lauren Forristal, ‘Hinge’s New AI Feature Determines If Your Prompt Response Is Too Basic’ (*TechCrunch*, 15 January 2025) <<https://techcrunch.com/2025/01/15/hinge-new-ai-feature-determines-if-your-prompt-response-is-too-basic/>> accessed 24 February 2025.

¹⁹⁹ ‘Create Posts with Our AI-Powered Writing Tool’ (*LinkedIn Help*) <<https://www.linkedin.com/help/linkedin/answer/a1517763>> accessed 24 February 2025.

²⁰⁰ Will Knight, ‘LinkedIn Is Testing an AI Tool That Could Transform How People Search for Jobs’ *Wired* <<https://www.wired.com/story/linkedin-job-search-artificial-intelligence/>> accessed 24 February 2025.

services²⁰¹, but such a policy is unlikely to absolve the company of its obligations under EU legislation.

These two faces turn VLOPs into entities wearing several hats under the AI Act. Broadly speaking, VLOPs that develop their own GPAI models and/or integrate them within their systems in any of the two mentioned faces are acting as providers. Additionally, when VLOPs implement AI systems for content moderation or recommendations, they act as ‘deployers’, which the AI Act defines as anyone who uses an AI system under their authority, unless it is used in a non-professional context.²⁰² These types of AI systems do not qualify as ‘high-risk’ under the AI Act. Moreover, the internal faces of GPAI models within the platform’s AI recommender and moderation systems would not qualify these as GPAI systems due to the specific nature of their task. In contrast, the external face would, in some cases, be much more general in nature, and have a greater potential to be classified as GPAI systems. Whether the models applied can be classified as GPAI models with systemic risk depends on their technical capacity.

Both uses of GPAI are part of the operationalization of VLOPs’ services. As was mentioned briefly in the previous section, the DSA requires VLOPs to implement risk assessments and mitigation strategies in relation to ‘the design or functioning of their service and its related systems, including algorithmic systems.’²⁰³ These risk assessments must identify possible systemic risks within their services, including the dissemination of illegal content; actual or foreseeable negative effects on fundamental rights or on civil discourse, electoral processes, public security, or on issues like gender-based violence, public health, the rights of minors, and the physical and mental well-being of individuals.²⁰⁴ When assessing such risks, the DSA provides that VLOPs must consider factors relating to the integration of GPAI models into their operations. These include the design of recommender systems and ‘any other relevant algorithmic system’,²⁰⁵ content moderation systems,²⁰⁶ advertising selection and presentation systems,²⁰⁷ and data practices within the platform.²⁰⁸ When risks are identified, the DSA requires platforms to implement mitigation strategies which can include adapting the relevant services or systems in light of the identified problems.

The first relevant interaction between the AI Act and the DSA is that the documentation required by the former from providers of GPAI models and those with systemic risks will be useful for risk assessments and mitigation strategies under the latter. For example, a platform implementing a GPAI model in its recommendation systems will clearly benefit from the technical information required by the AI Act on the training, testing and evaluation of such models²⁰⁹ to inform its assessment of whether its recommender systems are likely to favour discriminatory or hateful content. Likewise, if a platform implements its flagship GPAI model as a freely accessible chatbot and the model meets the technical criteria for being considered as one with systemic risk, AI Act’s requirement for model evaluations using adversarial

²⁰¹ Ellen Jennings-Trace published, ‘LinkedIn Says If You Share Fake or False AI-Generated Content, That’s on You’ (*TechRadar*, 9 October 2024) <<https://www.techradar.com/pro/linkedin-says-if-you-share-fake-or-false-ai-generated-content-thats-on-you>> accessed 25 February 2025.

²⁰² AI Act, art 3(4).

²⁰³ DSA, art 34(1).

²⁰⁴ *ibid.*

²⁰⁵ *ibid.*, art 34(2)(a).

²⁰⁶ *ibid.*, art 34(2)(b).

²⁰⁷ *ibid.*, art 34(2)(d).

²⁰⁸ *ibid.*, art 34(2)(e).

²⁰⁹ AI Act, art 53(1)(a).

testing²¹⁰ can help identify whether this new service generates sexist content, or content that could lead to self-harm, enabling the necessary modifications to be made.

Beyond these general requirements relating to risk assessment and management, the DSA has specific requirements that affect how GPAI is used on platforms.

7.1.1. The internal faces: VLOPs use of GPAI for recommending and moderating content

Recommender systems, which the DSA defines as those that suggest or prioritise information for platform users,²¹¹ are central to social media businesses and also have a significant democratic impact. The more accurate and relevant the recommendations or suggestions made by social media platforms are, the more attractive they are to users. Much of TikTok's growing success, for example, is attributed to the platform's 'impressive accuracy' in recommendations, which is often associated with users' alleged 'addiction' to the app.²¹² However, recommender systems can also contribute to the spread of harmful content that undermines rights and democracy.²¹³ In this regard, allegations have been made that TikTok's recommender system played a critical role in the victory of the ultranationalist candidate Călin Georgescu in the 2024 Romanian elections.²¹⁴ The Romanian Constitutional Court later annulled these elections based on intelligence documents revealing alleged Russian interference in the online discourse surrounding the vote.²¹⁵ Following this, the Commission initiated proceedings against TikTok regarding its risk management in the context of elections, with a particular focus on political advertising, paid political content and recommender systems.²¹⁶

The DSA does not impose specific restrictions on the use of AI in recommender or content moderation systems. Instead, it establishes measures to ensure fairness and transparency in the deployment of such systems.

In this respect, the DSA requires platforms to implement transparency measures regarding their recommender systems. These transparency measures do not include an obligation to disclose the technology used, but instead focus on requiring the platforms to include information in their terms and conditions about the parameters used in their systems to rank content suggested to users.²¹⁷ Platforms are also required to offer at least one recommender system option that does not rely on profiling,²¹⁸ to provide clear information about all available options, and to ensure users can easily and directly switch between them.²¹⁹

²¹⁰ AI Act, art 55(1)(a).

²¹¹ DSA, art 3(s).

²¹² Maximilian Boeker and Aleksandra Urman, 'An Empirical Investigation of Personalization Factors on TikTok', *Proceedings of the ACM Web Conference 2022* (ACM 2022) <<https://dl.acm.org/doi/10.1145/3485447.3512102>> accessed 26 May 2025.

²¹³ Judit Bayer, 'The Place of Content Ranking Algorithms on the AI Risk Spectrum' (2024) 48 *Telecommunications Policy* 102741.

²¹⁴ 'What Happened on TikTok around the Romanian Elections?' | Global Witness' <<https://globalwitness.org/en/campaigns/digital-threats/what-happened-on-tiktok-around-the-annulled-romanian-presidential-election-an-investigation-and-poll/>> accessed 26 May 2025.

²¹⁵ Alina Carrozzini, 'Shooting Democracy in the Foot?' (*Verfassungsblog*, 13 December 2024) <<https://verfassungsblog.de/shooting-democracy-in-the-foot/>> accessed 29 March 2025.

²¹⁶ 'Commission Opens Formal Proceedings against TikTok under DSA' (*European Commission - European Commission*) <https://ec.europa.eu/commission/presscorner/detail/en/ip_24_6487> accessed 26 May 2025.

²¹⁷ DSA, art 27(2).

²¹⁸ *ibid*, art 27(3).

²¹⁹ *ibid*, art 38.

In terms of transparency regarding the use of AI in moderation systems, the DSA only requires platforms to: i) disclose whether automated tools were used when responding to user notices of illegal content;²²⁰ ii) include information on the use of automated tools within the statements of reasons explaining restrictions on content or accounts;²²¹ and ii) include statistics on the use of such tools in their annual transparency reports, along with qualitative descriptions, information on their purposes, and details on their accuracy and error rates.²²² This is coupled with an obligation to allow users to have access to an internal appeals mechanism operating ‘under the supervision of appropriately qualified staff’.²²³

Since these two types of AI use within platforms do not classify as ‘high-risk’, the AI Act has hardly impacted their operationalization. Nevertheless, it is reasonable to assume that the different requirements for GPAI model providers will still influence how these systems are deployed. On the one hand, if platforms implement GPAI models, the transparency requirements of the DSA could be combined with the documentation requirements for such models under the AI Act.

In that sense, it should be noted that while the annual report required by the DSA calls for information on qualitative descriptions, purposes and details of accuracy and error rates, it does not require the disclosure of the specific technology being used. In practice, the first reports from some of the VLOPs are limited to vague terms such as ‘neural networks’, ‘computer vision’ and ‘natural language processing’.²²⁴ Likewise, the requirement to disclose the ‘parameters’ used by recommender algorithms, as well as their relevance, does not refer to technical specifications. Rather than requiring platforms to disclose the AI tools used to generate recommendations, the DSA asks for disclosure of the types of information processed by these tools and the processing instructions.

Against that backdrop, a meaningful application of the documentation requirements imposed by the AI Act on GPAI model providers within this context would be to interpret them as enhancing transparency obligations where recommender and moderation systems rely on such models. For example, the information on training, testing, and evaluation processes required by the AI Act²²⁵ would significantly complement the data that platforms must include in their annual reports on automated moderation under the DSA. Similarly, users would likely gain a deeper understanding of recommender systems based on GPAI models if platforms published the training data information mandated for disclosure by the AI Act.²²⁶

Furthermore, the requirements imposed on providers of GPAI models with systemic risk can be seen as enhancing the safety of recommender and moderation systems that use these models. In this context, the AI Act’s requirements for conducting model evaluations,²²⁷ assessing and mitigating risks,²²⁸ documenting and reporting serious incidents²²⁹ and ensuring

²²⁰ *ibid*, art 16(6).

²²¹ *ibid*, art 17(3)(c).

²²² *ibid*, arts 15 and 42.

²²³ *ibid*, art 20(6).

²²⁴ Vargas Penagos (n 194).

²²⁵ AI Act, art 53(1)(a).

²²⁶ AI Act, art 53(1)(d).

²²⁷ AI Act, art 55(1)(a).

²²⁸ AI Act, art 55(1)(b).

²²⁹ AI Act, art 55(1)(c).

cybersecurity²³⁰ would establish a baseline of safety requirements for GPAI models used for such tasks.

Since platforms in this context would be acting as ‘deployers’ of AI systems, they would be bound by the obligation set out in the AI Act to ensure that staff operating those tools are literate.²³¹ This would enhance the DSA requirement for appeals mechanisms to be ‘under the supervision of appropriately qualified staff,’²³² as well as setting a baseline for the technical training of personnel involved in moderation and recommendations.

Finally, platforms that use GPAI models to recommend or moderate content would typically not face liability for the content they allow to be disseminated. This is because, as Melian Pérez explains, platforms’ control over the models is limited to technical aspects.²³³

7.1.2. The external face: VLOPs’ use of GPAI as a service

Platforms that incorporate GPAI models as services for users, as mentioned at the beginning of this section, would act as providers under the AI Act. This is because they either develop an AI system themselves or have one developed by a third party and then put it into service under their own name or trademark. In some cases, the service provided may also be considered as a GPAI system. An example of this could be X’s Grok, which generates text and images and interacts directly with users, responding to queries or questions when mentioned in a post.²³⁴ However, services such as those deployed by LinkedIn for post generation²³⁵ or advanced job searches²³⁶ would be much more specific in nature.

However, the various facets of services like Grok in X warrant particular attention. Not only does Grok have an interface through which users can make prompts to receive replies, it also interacts with users through its own user account.²³⁷ The latter raises questions as the chatbot itself becomes part of the user-generated content flow. Up until June 2025, X’s rules and policies,²³⁸ privacy policy²³⁹ and terms of service²⁴⁰ do not specify whether there are particular rules or channels for moderating Grok’s posts. Furthermore, X’s page explaining Grok provides limited information on its training, and how people can opt out, but says nothing about moderation.²⁴¹

This is a critical issue because large language models such as those underlying Grok, ChatGPT and other similar GPAI-based chatbots can produce responses that sound

²³⁰ AI Act, art 55(1)(d).

²³¹ AI Act, art 4.

²³² DSA, art 20(6).

²³³ Gabriel Ernesto Melian Melian Pérez, ‘From Curators to Creators: Navigating Regulatory Challenges for General-Purpose Generative AI in Europe’ (2025) 16 JIPITEC – Journal of Intellectual Property, Information Technology and E-Commerce Law <<https://www.jipitec.eu/jipitec/article/view/437>> accessed 8 September 2025.

²³⁴ ‘Elon Musk’s Startup Rolls out New Grok-3 Chatbot as AI Competition Intensifies’ (n 196); ‘X Now Lets You Mention Grok in Replies, Get Explanations’ (n 196).

²³⁵ ‘Create Posts with Our AI-Powered Writing Tool’ (n 199).

²³⁶ Knight (n 200).

²³⁷ ‘Elon Musk’s Startup Rolls out New Grok-3 Chatbot as AI Competition Intensifies’ (n 195); ‘X Now Lets You Mention Grok in Replies, Get Explanations’ (n 195).

²³⁸ ‘The X Rules: Safety, Privacy, Authenticity, and More’ <<https://help.x.com/en/rules-and-policies/x-rules>> accessed 6 June 2025; ‘Rules and Policies’ <<https://help.x.com/en/rules-and-policies>> accessed 6 June 2025.

²³⁹ ‘X Privacy Policy’ <<https://x.com/en/privacy>> accessed 6 June 2025.

²⁴⁰ ‘X Terms of Service’ <<https://x.com/en/tos>> accessed 6 June 2025.

²⁴¹ ‘About Grok’ <<https://help.x.com/en/using-x/about-grok>> accessed 6 June 2025.

convincing but are not based on facts.²⁴² An example of these emerging challenges is the controversy surrounding Grok's malfunction in May 2025, when the bot provided repeated, unrelated responses to a large number of users that included endorsements of the 'white genocide' conspiracy theory in South Africa. X's owner, Elon Musk, attributed this to an unauthorised tweak by someone within the company.²⁴³ This incident demonstrates how minor factors can pose risks to public discourse. It should also be noted that Grok's descriptive webpage states that this chatbot has 'access to real-time public X posts' in order to provide 'up-to-date information and insights on a wide range of topics.'²⁴⁴ Fact-checking organisations have raised concerns about how people are now using this feature as an AI fact-checking tool, which carries a high risk of spreading misinformation.²⁴⁵ As it is fed with real-time public X posts, Grok could be prone to replicating unverified rumours, misinformation, disinformation, propaganda and hate speech that are being spread in 'real-time' and are prominent in public discourse.²⁴⁶ These issues also raise questions about the liability of platforms for content generated by such services. Their role as intermediaries shifts significantly in these contexts. Although publications are still triggered by user prompts, platforms may also exert influence and exercise control over the final output depending on different technical factors such as how much the model replicates existing content or how much it abstracts it to create a new one.²⁴⁷ Moreover, given that the PLD 2024 does not apply to 'information', it would be an inadequate framework for these situations.

In most cases, platforms deploying GPAI-based services would, in most cases, not fall under the high-risk category. As a result, they would generally be permitted to implement voluntary measures, primarily in the form of codes of conduct.²⁴⁸ If their services are intended to interact directly with users, they must be developed in a way that enables users to discern whether they are interacting with a human or not.²⁴⁹ For instance, if Meta were to continue with its plans to create AI-generated profiles,²⁵⁰ the AI Act would require them to make it clear to users that they are communicating with a chatbot. Moreover, if they are providing a service that generates synthetic audio, text, video, or images, they would be required to implement machine-readable identifiers.²⁵¹ Similarly, when Grok is used as a user account on social media services, the AI Act's application of this rule should be understood as requiring disclaimers in its replies about the potential for errors. Furthermore, if Grok's underlying model were to be classified as a GPAI system with systemic risk, X would be expected to address this in the model's risk assessment and mitigation.

Furthermore, such risks are highly likely to be covered by the DSA's obligations for VLOPs, as they can clearly impact the dissemination of illegal content and the exercise of fundamental

²⁴² G Pradeep Reddy, YV Pavan Kumar and K Purna Prakash, 'Hallucinations in Large Language Models (LLMs)', *2024 IEEE Open Conference of Electrical, Electronic and Information Sciences (eStream)* (IEEE 2024) <<https://ieeexplore.ieee.org/document/10542617/>> accessed 19 July 2024.

²⁴³ 'Musk's xAI Blames "unauthorized" Tweak for "White Genocide" Posts' (*France 24*, 17 May 2025) <<https://www.france24.com/en/live-news/20250517-musk-s-xai-blames-unauthorized-tweak-for-white-genocide-posts>> accessed 6 June 2025.

²⁴⁴ 'About Grok' (n 241).

²⁴⁵ Jagmeet Singh, 'X Users Treating Grok like a Fact-Checker Spark Concerns over Misinformation' (*TechCrunch*, 19 March 2025) <<https://techcrunch.com/2025/03/19/x-users-treating-grok-like-a-fact-checker-spark-concerns-over-misinformation/>> accessed 6 June 2025.

²⁴⁶ 'Hey Chatbot, Is This True? AI "factchecks" Sow Misinformation' (*France 24*, 2 June 2025) <<https://www.france24.com/en/live-news/20250602-hey-chatbot-is-this-true-ai-factchecks-sow-misinformation>> accessed 6 June 2025.

²⁴⁷ Melian Pérez (n 233).

²⁴⁸ AI Act, art 95.

²⁴⁹ AI Act, art 50(1).

²⁵⁰ Morrow (n 197).

²⁵¹ AI Act, art 50(2).

rights—for instance, by perpetuating discriminatory stereotypes—as well as influencing elections and public health.²⁵² Several of the DSA’s mitigation strategies could apply here, such as ‘adapting the design, features or functioning of their services, including their online interfaces,’²⁵³ ‘adapting their terms and conditions and their enforcement,’²⁵⁴ or ‘adapting any relevant decision-making processes and dedicated resources for content moderation.’²⁵⁵ Notably, these strategies are not limited to the fixed one-year period, as Article 34(1) establishes that risk assessments should also be conducted prior to the deployment of functionalities ‘likely to have a critical impact on the risks identified’. This obligation is becoming increasingly relevant given the speed with which GPAI services are being developed and deployed by social media platforms. As mentioned previously, Recital 118 states that VLOPs should apply the risk-framework of the DSA, while the obligations of the AI Act would be activated if ‘significant systemic risks’ not covered by the former are identified in the AI technologies incorporated in the platforms’ services. According to the Recital, VLOPs must therefore assess systemic risks, taking into account ‘how the design of algorithmic systems used in the service may contribute to’ these risks.’

8. Evaluation: Where are responsibility and liability best allocated in the GPAI value chain?

So far, this article has provided a comprehensive explanation of how the AI Act, the GPSR, the PLD 2024 and the DSA together form an extensive legal framework that may be relevant for allocating responsibility and liability in the GPAI value chain.

In practice, model providers usually have the best understanding of how their models work, while downstream providers are better placed to gather information on the real-world risks and impacts of these technologies. Deployers, in turn, may have the greatest capacity to implement mitigation measures and monitor system performance. However, the further down the value chain we go, the less likely it is that detailed information about how the technology works will be available.²⁵⁶ While the different layers of the value chain are highly interdependent through information sharing and documentation, the non-linear and dynamic nature of the GPAI value chain complicates allocation of responsibility and liability.²⁵⁷

8.1. Upstream vis-a-vis downstream, navigating the whirlpool

Articles 53 and 55 of the AI Act set out the relevant rules for GPAI model providers with regard to transparency, documentation, risk assessment and mitigation, and the provision of documentation to enable downstream development and compliance. As this article has shown, however, downstream providers in the GPAI value chain bear heavier duties only when they integrate a GPAI model into a high-risk AI system, or when they substantially modify a GPAI system in a way that brings it into the high-risk category. Without adequate cooperation

²⁵² DSA, art 34(1).

²⁵³ *ibid*, art 35(1)(a).

²⁵⁴ *ibid*, art 35(1)(b).

²⁵⁵ *ibid*, art 35(1)(c).

²⁵⁶ Engler and Renda (n 30).

²⁵⁷ *ibid*.

between both streams of the value chain, the safeguards introduced by Articles 53 and 55 of the AI Act are little more than window dressing.

In this context, Article 25(4) of the AI Act is crucial in establishing that model providers must agree with high-risk downstream providers on the information and other factors that are necessary for the latter to comply with the conformity requirements of that regulation. However, this provision does not take into account the fact that model providers will usually be powerful actors with an interest in keeping information that they consider to be a trade secret confidential.

In terms of liability, the PLD 2024 focuses on the final product. This is evident in the language used to assess defectiveness, which refers to the moment a product is ‘placed on the market or put into service.’ Similarly, its broad definition of ‘manufacturer’, which includes anyone who develops, manufactures, or produces a product, as well as anyone presenting themselves as its manufacturer, also directs attention to the final product. Furthermore, although Article 12(2) of the Directive permits agreements between component manufacturers (i.e., model providers) and product manufacturers whereby the latter to waive their right of recourse against the former, Article 12(1) still requires both to be held jointly and severally liable. In practice, this means that users can seek redress from either or both parties in the value chain. However, it should be noted that this Directive does not address liability by AI deployers, who may have a more direct link to injured parties. In such situations, liability will fall under broader contractual or non-contractual liability regimes, without prejudice to the liability of upstream or downstream providers.

Moreover, cooperation between upstream and downstream providers is essential for liability purposes because the PLD 2024 considers compliance with safety requirements when evaluating a product’s defectiveness.

Finally, the Commission’s introduction of the threshold for determining a ‘substantial modification’ of a GPAI model is relevant in determining the role of model modifiers in terms of responsibility and liability. If they surpass the threshold, they must comply with Article 53 of the AI Act (and eventually with Article 55). At the same time, they would be component manufacturers, which could lead to joint and several liability in case of defectiveness.

8.2. The question of contractual freedom

Some major model providers seek to limit their liability for model outputs through contracts.²⁵⁸ Examples of this can be seen in the B2B terms of service of companies such as Anthropic,²⁵⁹ OpenAI²⁶⁰ and X.AI.²⁶¹

²⁵⁸ *ibid.*

²⁵⁹ ‘the liability of each party, and its affiliates and licensors, for any damages arising out of or related to these Terms (i) excludes damages that are consequential, incidental, special, indirect, or exemplary damages, including lost profits, business, contracts, revenue, goodwill, production, anticipated savings, or data, and costs of procurement of substitute goods or services and (ii) is limited to Fees paid by Customer for the Services in the previous 12 months.’ See: Anthropic, ‘Commercial Terms of Service’ <<https://www.anthropic.com/legal/commercial-terms>> accessed 11 August 2025.

²⁶⁰ ‘Customer is solely responsible for all use of the Outputs and for evaluating the accuracy and appropriateness of Output for Customer’s use case.’ See: OpenAI, ‘Open AI Service Agreement’ <<https://cdn.openai.com/osa/openai-services-agreement.pdf>> accessed 8 August 2025.

²⁶¹ ‘Customer understands and agrees that the Output may contain “hallucinations” and may be inaccurate, objectionable, inappropriate, or otherwise unsuited to Customer’s purpose, and Customer agrees that xAI shall not be liable for any damages

This raises the question of whether it is possible to review the content of contractual terms that disproportionately or unfairly tilt the balance of responsibility and liability along the value chain. This possibility is challenging in practice because many legal systems do not have mechanisms for such reviews, and those that do often do not allow reviews of clauses related to the main subject matter of the contract. Arguably, clauses related to the unreliability of outputs fall within that category.

An extreme solution would be to prohibit liability waivers, as was considered during the legislative process for the upcoming New York legislation on this matter.²⁶² However, such an approach would be hard to justify in the EU because it would conflict with safeguards for contractual freedom and the freedom to conduct a business. On the other end of the spectrum, the most lenient approach would be to leave the issue entirely unrestricted, but this would carry the risk of unfairness.

Against this backdrop, several EU digital laws make B2B agreements or terms of service subject to review in light of the FRAND principles (fair, reasonable and non-discriminatory). More specifically, after the CJEU applied these principles within the context of Article 102 TFEU,²⁶³ they have been incorporated into EU secondary legislation in the digital domain as benchmarks for assessing the fairness of contractual terms.²⁶⁴ This approach is reflected, for instance, in the Digital Markets Act (DMA),²⁶⁵ the Data Act (DA)²⁶⁶ and the Data Governance Act.²⁶⁷

Such waiver clauses may arguably be contrary to FRAND principles in some situations, as they place the liability burden on a party that may not be best placed to fully mitigate or foresee the harmful outcomes of the final product arising from a model's defectiveness. Moreover, they would likely conflict with Article 12(1) of the PLD 2024, which establishes joint and severable liability for product and component manufacturers.

Similar challenges arise in the context of public procurement, particularly when government agencies become embedded as downstream developers in the value chain rather than merely deployers. While these entities are expected to hold significant market power, the provider may actually operate in a concentrated market with greater knowledge and capabilities. Consequently, the agency's operational needs may cause it to concede to the provider's demands.²⁶⁸ In the absence of horizontal legislation for the public procurement of AI systems,

Customer or any third party alleges to incur as a result of or relating to any Output or other content generated by or accessed on or through the Service.' See: X.ai, 'Terms of Service - Enterprise | xAI' <<https://x.ai/legal/terms-of-service-enterprise>> accessed 11 August 2025.

²⁶² RAISE act (n 91).

²⁶³ See for instance Case C-170/13 *Huawei Technologies Co Ltd v ZTE Corp, ZTE Deutschland GmbH* EU:C:2015:477. Mathew Heim and Igor Nikolic, 'A FRAND Regime for Dominant Digital Platforms' (2019) 10 JIPITEC – Journal of Intellectual Property, Information Technology and E-Commerce Law 38.

²⁶⁴ See Michael Denga and Samuel Walsh, 'Fair, reasonable and non-discriminatory' – KI-Dienstleistungen, Vertragsrecht und die FRAND-Prinzipien' (2025) 6 *JuristenZeitung* 243.

²⁶⁵ Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector and amending Directives (EU) 2019/1937 and (EU) 2020/1828 [2022] OJ L 265, 12.10.2022, pp. 1–66, art 6(12).

²⁶⁶ Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonised rules on fair access to and use of data and amending Regulation (EU) 2017/2394 and Directive (EU) 2020/1828 [2023], arts 8(1) and 13(3).

²⁶⁷ Regulation (EU) 2022/868 of the European Parliament and of the Council of 30 May 2022 on European data governance and amending Regulation (EU) 2018/1724 [2022] OJ L 152, 3.6.2022, pp. 1–44, art 12(f).

²⁶⁸ Albert Sanchez-Graells, 'Public Procurement of Artificial Intelligence: Recent Developments and Remaining Challenges in EU Law' [2024] SSRN Electronic Journal <<https://www.ssrn.com/abstract=4706400>> accessed 12 August 2025.

the Commission has developed a set of model contractual clauses, which were recently updated in February 2025.²⁶⁹ These model clauses are, however, limited to the context of government agencies buying systems to act as deployers, with an emphasis on high-risk AI systems. For non-high-risk AI systems, the model clauses provide a ‘light version’ aimed at meeting basic transparency and explainability standards in public administration, while being less strict than the requirements for high-risk systems.²⁷⁰

8.3. No fixed stream

As emphasised previously, responsibility and liability in the GPAI value chain should be allocated according to each actor’s ability to detect, prevent and mitigate harm. Model providers are generally best placed to ensure transparency, documentation, and technical safeguards relating to how the model works. In contrast, downstream providers and deployers are better positioned to provide information relevant to assessing real-world risks and impacts, and to mitigate them. Effective allocation requires strong cooperation between upstream and downstream actors, as well as clear contractual agreements that support such cooperation. Moreover, although the liability framework may protect the end user by enabling joint and several liability, allowing contractual agreements for waiving the right to recourse by downstream providers²⁷¹ may concentrate power in the hands of model providers. In practice, however, information asymmetries, market concentration, and imbalances in bargaining power can distort this allocation, highlighting the need for regulatory oversight and fair contractual frameworks.

Finally, and as can be seen in the case of VLOPs, integrating GPAI models in different industries raises additional considerations arising from sector-specific legislation. While the AI Act is likely to increase transparency obligations for these platforms, it does not clarify the liability they may face for misuse or harmful outcomes.

8.4. In between

Model marketplaces could play a crucial role in preventing the dissemination of models with a higher likelihood of misuse or harmful outcomes. One possible solution would be to adopt a stricter interpretation of their obligations as hosting providers under the DSA. This could involve lowering the threshold for what constitutes ‘actual knowledge’ and introducing monitoring duties. However, such an approach would be costly. Given the technical complexity of assessing each model, it could also discourage the existence of such marketplaces altogether. Research on intermediary liability shows that strict regimes often lead to ‘overmoderation’ or ‘overcompliance’. Platforms face risks associated with harmful content, but not the benefits of harmless expression. Since they have imperfect information, they tend

²⁶⁹ ‘Updated EU AI Model Contractual Clauses Now Available | Public Buyers Community’ (5 March 2025) <<https://public-buyers-community.ec.europa.eu/communities/procurement-ai/news/updated-eu-ai-model-contractual-clauses-now-available>> accessed 12 August 2025.

²⁷⁰ Public Buyers Community, ‘Commentary - Model Contractual Clauses for the Public Procurement of AI (MCC-AI) Procurement of AI | Public Buyers Community’ <<https://public-buyers-community.ec.europa.eu/system/files/2025-05/Commentary.pdf>> accessed 12 August 2025.

²⁷¹ PLD, art 12(2)(b).

to remove more content than is socially or legally required in order to avoid liability or costly reviews.²⁷²

Against that backdrop, the AI Office could play a proactive role by encouraging the adoption of further Codes of Practice that involve the cooperation of model marketplaces within the value chain. This could include, for example, the minimum amount of information that upstream providers must provide when uploading models to these marketplaces.

9. Conclusions

This article has demonstrated that the approach adopted by the AI Act, along with complementary instruments such as the Code of Practice and the Commission guidelines, falls short of capturing the complexities stemming from the fluid and dynamic nature of the GPAI value chain. While the AI Act introduces relevant provisions, such as certain cooperation duties, its approach to risk management is inconsistent, focusing primarily on GPAI model providers. As discussed throughout the article, however, downstream providers play a significant role in risk mitigation and assessment. Although these are tasks that upstream providers are legally expected to undertake, downstream providers are in a better position to assess real-world risks. The AI Act also overlooks the critical role of model marketplaces in disseminating models.

Some of these concerns are addressed by additional legislation, including the GPSR, the PLD 2024, and the DSA. However, there does not appear to be a clear strategy from the relevant authorities regarding how these frameworks might work together or how potential loopholes should be closed. In that regard, it is worth noting that the Commission is currently undergoing a process of proposing a simplification package.²⁷³ This may be an opportunity to address some of the complications emerging from the current legislation.

A more coherent and integrated approach would therefore be desirable. Such an approach would enable the framework to reflect the distributed nature of risks more effectively than by predominantly concentrating obligations on model providers. However, this would require an assessment of what is technically feasible: an issue that calls for further research.

²⁷² James Grimmelman and Pengfei Zhang, 'An Economic Model of Online Intermediary Liability' (2023) 38 Berkeley Technology Law Journal 1011.

²⁷³ European Parliament, 'Digital Package | Legislative Train Schedule' (*European Parliament*) <<https://www.europarl.europa.eu/legislative-train/theme-a-new-plan-for-europe-s-sustainable-prosperity-and-competitiveness/file-digital-package>> accessed 20 October 2025.

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